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TOPIC – BREAST CANCER DIAGNOSIS: A LOGISTIC REGRESSION-BASED APPROACH

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EXECUTIVE SUMMARY

This project work focuses on building and evaluating a logistic regression model to predict breast cancer occurrences. Using a dataset consisting of cytological features obtained from Fine Needle Aspirate (FNA) tests, the model aims to estimate the likelihood of malignancy in breast tumours. Here 10 predictors are considered such as radius, texture, smoothness, compactness, concavity, symmetry, fractal dimension etc. From literature these variables are known to be associated with cancerous growth.

The process involves multiple stages, including data preprocessing, feature selection, model training, and performance evaluation. The objective is to develop a reliable and interpretable logistic model that can support early diagnosis and may help in clinical decision-making.

The present project work fulfils it's objective by providing a well-fitted model that enables early diagnosis of malignant breast cancer.

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1. INTRODUCTION

Breast cancer is one of the most common cancers affecting women worldwide. It occurs when cells in the breast grow uncontrollably, potentially forming malignant tumours that can spread if not detected early. The ability to distinguish between benign and malignant tumours is crucial in guiding timely treatment and improving patient outcomes.

The data used comes from routine diagnostic tests, specifically fine-needle aspirate (FNA) results. These tests measure features like the shape, size, and texture of cell nuclei. Using this information, we aim to train a logistic regression model that can learn patterns from past cases and make predictions on new ones.

With the growing availability of medical data, statistical models have become valuable tools in supporting clinical decision-making. Among these, logistic regression has proven effective in predicting binary outcomes, such as the presence or absence of disease, based on measurable factors.

By analysing characteristics of cell structures obtained through diagnostic tests, logistic regression can estimate the likelihood of a tumour being malignant. Its strength lies in its simplicity, interpretability, and usefulness in medical settings where transparency in prediction is important.

This project explores the potential of logistic regression to aid in the diagnosis of breast cancer, using real-world medical data. It reflects the broader goal of integrating data-driven approaches into healthcare, where even simple models can help inform decisions and potentially improve the accuracy of early cancer detection.

2. CAUTION

Due to ethical concerns and the inherent limitations of machine learning in medical diagnosis, this model should not be relied upon as the sole method for detecting breast cancer. Instead, it may serve as an initial screening aid to support and prompt further clinical assessment by qualified medical professionals.

3. ASSUMPTIONS

➡ Here are the key assumptions for Logistic regression-

I. **Binary Outcome:**

Binary logistic regression requires the dependent/response variable to be **dichotomous** in nature, that means the variable must be binary (e.g., 0/1, yes/no).

II. **Independence of observations:**

Each observation should be **independent** of the others.

III. **No multicollinearity:**

Logistic regression requires there to be little or **no multicollinearity** among the predictor/explanatory variables. This means that the independent variables must not be too highly correlated with each other.

IV. **Linearity of independent variables with log odds:**

There must be a **linear relationship** between continuous predictors and the logit (log-odds of the dependent variable) of the outcome.

V. **Large sample size:**

Logistic regression analysis yields reliable, robust, and valid results when a **larger sample size of the data-set** is considered.

4. DATA SOURCE

The original data was collected as part of a study conducted by **William Wolberg, Olvi Mangasarian, Nick Street, W. Street** around 1995.(The original archive: <https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>)

This refined BREAST CANCER dataset is collected from the website Kaggle.com. The link to the original data is attached below.

<https://www.kaggle.com/datasets/yasserh/breast-cancer-dataset>

5. DATA COLLECTION METHOD

In our data, features are extracted from a digitized image of a fine needle aspirate (FNA) of a breast mass using image processing technique. They describe characteristics of the cell nuclei present in the image.

➤ Sample collection via FNA (Fine Needle Aspiration)

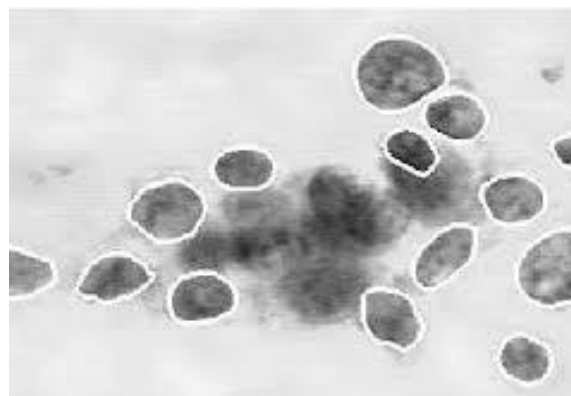
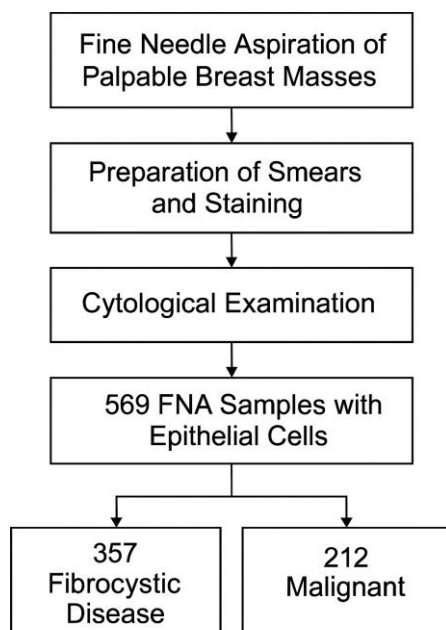
In this study, we analyzed 569 consecutive fine needle aspiration (FNA) samples from palpable, non-cystic breast masses, each containing epithelial cells. The method is given below:

- A **23-gauge needle** is inserted into the breast lump.
- **Negative pressure** is applied via a syringe to aspirate cells.
- The aspirated material is spread onto **silane-coated slides**.
- The slide is immediately **fixed and stained** (e.g., Papanicolaou stain).

➤ Image Acquisition

After Fine Needle Aspiration (FNA), the area on the aspirate slide selected for analysis was visually chosen to ensure minimal nuclear overlap, which is essential for accurate cytological evaluation-

- The prepared slide is placed under a **microscope** (e.g., Olympus).
- A **video camera** (e.g., JVC TK-1070U) is mounted on the microscope.
- A **frame grabber** (e.g., ComputerEyes/RT) captures the image digitally.
- Magnification used: e.g., **63x objective and 2.5x ocular lens**.



A magnified image of a malignant breast FNA

(Figure from Mangasarian OL, Street WN., Wolberg, WH. Breast Cancer Diagnosis and Prognosis via Linear Programming. Mathematical Programming Technical Report 94-10. 1994 Dec)

6. DATA DESCRIPTION

The computer vision diagnostic system extracts ten nuclear features from the isolated cell nuclei. All of the features are numerically modeled such that larger values indicate a higher likelihood of malignancy. The mean value, worst (mean of the three largest values), and standard error of each feature were computed for each image, resulting in a total of thirty features for each case in the study along with the response variable diagnosis. The value of a feature for an image is the mean of the values for the individual nucleus

Diagnosis: The diagnosis of the tumor, where M stands for malignant and B stands for benign.

1. **Radius** – Mean of distances from center to points on the perimeter.
2. **Texture** – Standard deviation of gray-scale values (represents roughness).
3. **Perimeter** – Length of the boundary of the nucleus.
4. **Area** – Size of the nucleus (number of pixels within the boundary).
5. **Smoothness** – Local variation in radius lengths.
6. **Compactness** – $(\text{Perimeter}^2 / \text{Area}) - 1$; describes how tightly packed the cell is.
7. **Concavity** – Severity of concave portions of the contour.
8. **Concave points** – Number of concave portions of the contour.
9. **Symmetry** – Symmetry of the nucleus.
10. **Fractal dimension** – Measurement of the complexity of the contour.

7. DATA PRE-PROCESSING

Data pre-processing refers to the process of transforming raw data into a format suitable for analysis, including tasks like cleaning, transforming, and integrating it to ensure consistency and accuracy.

7.1) Initial sanity checking for data analysis:

An initial sanity check is a quick validation process to ensure a claim, calculation, or proposed functionality is plausible and free from obvious errors or flaws. The main goal of a sanity check is to rule out obviously incorrect or absurd results or assumptions. It's not meant to catch every possible error, but rather to identify gross mistakes quickly. The sanity checking process includes the following points-

7.1.1 Data Dimension

```
> dim(Data)
[1] 569 31
```

7.1.2 Variable names

```
> names(Data)
[1] "diagnosis"          "radius_mean"      "texture_mean"
[4] "perimeter_mean"     "area_mean"        "smoothness_mean"
[7] "compactness_mean"   "concavity_mean"    "concave.points_mean"
[10] "symmetry_mean"      "fractal_dimension_mean" "radius_se"
[13] "texture_se"         "perimeter_se"     "area_se"
[16] "smoothness_se"      "compactness_se"    "concavity_se"
[19] "concave.points_se"  "symmetry_se"       "fractal_dimension_se"
[22] "radius_worst"       "texture_worst"     "perimeter_worst"
[25] "area_worst"         "smoothness_worst"  "compactness_worst"
[28] "concavity_worst"    "concave.points_worst" "symmetry_worst"
[31] "fractal_dimension_worst"
```

7.1.3 Finding Missing Values

Before starting our analysis we need to check if there present any missing values in the dataset or not, then we proceed accordingly.

```
> which(is.na(Data))
integer(0)
```

So, we can conclude that there is no missing values present in the dataset.

7.2) Encoding:

As the variable “diagnosis” has character-type data in it so we need to make them numeric type.

We have “M” for Malignant and “B” for Benign. In order to analysis the data we convert “M” to “1” and “B” to “0”.

```
> Data$diagnosis[Data$diagnosis=="M"]<-1
> Data$diagnosis[Data$diagnosis=="B"]<-0
```

7.3) Data Summary:

The summary of each variable is given below -

```
> summary(Data)
diagnosis      radius_mean    texture_mean    perimeter_mean    area_mean
Min.   :0.0000   Min.   : 6.981   Min.   : 9.71   Min.   : 43.79   Min.   : 143.5
1st Qu.:0.0000   1st Qu.:11.700   1st Qu.:16.17   1st Qu.: 75.17   1st Qu.: 420.3
Median :0.0000   Median :13.370   Median :18.84   Median : 86.24   Median : 551.1
Mean   :0.3726   Mean   :14.127   Mean   :19.29   Mean   : 91.97   Mean   : 654.9
3rd Qu.:1.0000   3rd Qu.:15.780   3rd Qu.:21.80   3rd Qu.:104.10   3rd Qu.: 782.7
Max.   :1.0000   Max.   :28.110   Max.   :39.28   Max.   :188.50   Max.   :2501.0

smoothness_mean    compactness_mean    concavity_mean    concave.points_mean    symmetry_mean
Min.   :0.05263   Min.   :0.01938   Min.   :0.00000   Min.   :0.00000   Min.   :0.1060
1st Qu.:0.08637   1st Qu.:0.06492   1st Qu.:0.02956   1st Qu.:0.02031   1st Qu.:0.1619
Median :0.09587   Median :0.09263   Median :0.06154   Median :0.03350   Median :0.1792
Mean   :0.09636   Mean   :0.10434   Mean   :0.08880   Mean   :0.04892   Mean   :0.1812
3rd Qu.:0.10530   3rd Qu.:0.13040   3rd Qu.:0.13070   3rd Qu.:0.07400   3rd Qu.:0.1957
Max.   :0.16340   Max.   :0.34540   Max.   :0.42680   Max.   :0.20120   Max.   :0.3040

fractal_dimension_mean    radius_se    texture_se    perimeter_se    area_se
Min.   :0.04996   Min.   :0.1115   Min.   :0.3602   Min.   : 0.757   Min.   : 6.802
1st Qu.:0.05770   1st Qu.:0.2324   1st Qu.:0.8339   1st Qu.: 1.606   1st Qu.:17.850
Median :0.06154   Median :0.3242   Median :1.1080   Median : 2.287   Median :24.530
Mean   :0.06280   Mean   :0.4052   Mean   :1.2169   Mean   : 2.866   Mean   :40.337
3rd Qu.:0.06612   3rd Qu.:0.4789   3rd Qu.:1.4740   3rd Qu.: 3.357   3rd Qu.:45.190
Max.   :0.09744   Max.   :2.8730   Max.   :4.8850   Max.   :21.980   Max.   :542.200

smoothness_se    compactness_se    concavity_se    concave.points_se    symmetry_se
Min.   :0.001713   Min.   :0.002252   Min.   :0.00000   Min.   :0.000000   Min.   :0.007882
1st Qu.:0.005169   1st Qu.:0.013080   1st Qu.:0.01509   1st Qu.:0.007638   1st Qu.:0.015160
Median :0.006380   Median :0.020450   Median :0.02589   Median :0.010930   Median :0.018730
Mean   :0.007041   Mean   :0.025478   Mean   :0.03189   Mean   :0.011796   Mean   :0.020542
3rd Qu.:0.008146   3rd Qu.:0.032450   3rd Qu.:0.04205   3rd Qu.:0.014710   3rd Qu.:0.023480
Max.   :0.031130   Max.   :0.135400   Max.   :0.39600   Max.   :0.052790   Max.   :0.078950

fractal_dimension_se    radius_worst    texture_worst    perimeter_worst    area_worst
Min.   :0.0008948   Min.   : 7.93   Min.   :12.02   Min.   : 50.41   Min.   :185.2
1st Qu.:0.0022480   1st Qu.:13.01   1st Qu.:21.08   1st Qu.: 84.11   1st Qu.:515.3
Median :0.0031870   Median :14.97   Median :25.41   Median : 97.66   Median :686.5
Mean   :0.0037949   Mean   :16.27   Mean   :25.68   Mean   :107.26   Mean   :880.6
3rd Qu.:0.0045580   3rd Qu.:18.79   3rd Qu.:29.72   3rd Qu.:125.40   3rd Qu.:1084.0
Max.   :0.0298400   Max.   :36.04   Max.   :49.54   Max.   :251.20   Max.   :4254.0

smoothness_worst    compactness_worst    concavity_worst    concave.points_worst    symmetry_worst
Min.   :0.07117   Min.   :0.02729   Min.   :0.0000   Min.   :0.00000   Min.   :0.1565
1st Qu.:0.11660   1st Qu.:0.14720   1st Qu.:0.1145   1st Qu.:0.06493   1st Qu.:0.2504
Median :0.13130   Median :0.21190   Median :0.2267   Median :0.09993   Median :0.2822
Mean   :0.13237   Mean   :0.25427   Mean   :0.2722   Mean   :0.11461   Mean   :0.2901
3rd Qu.:0.14600   3rd Qu.:0.33910   3rd Qu.:0.3829   3rd Qu.:0.16140   3rd Qu.:0.3179
Max.   :0.22260   Max.   :1.05800   Max.   :1.2520   Max.   :0.29100   Max.   :0.6638

fractal_dimension_worst
Min.   :0.05504
1st Qu.:0.07146
Median :0.08004
Mean   :0.08395
3rd Qu.:0.09208
Max.   :0.20750
```

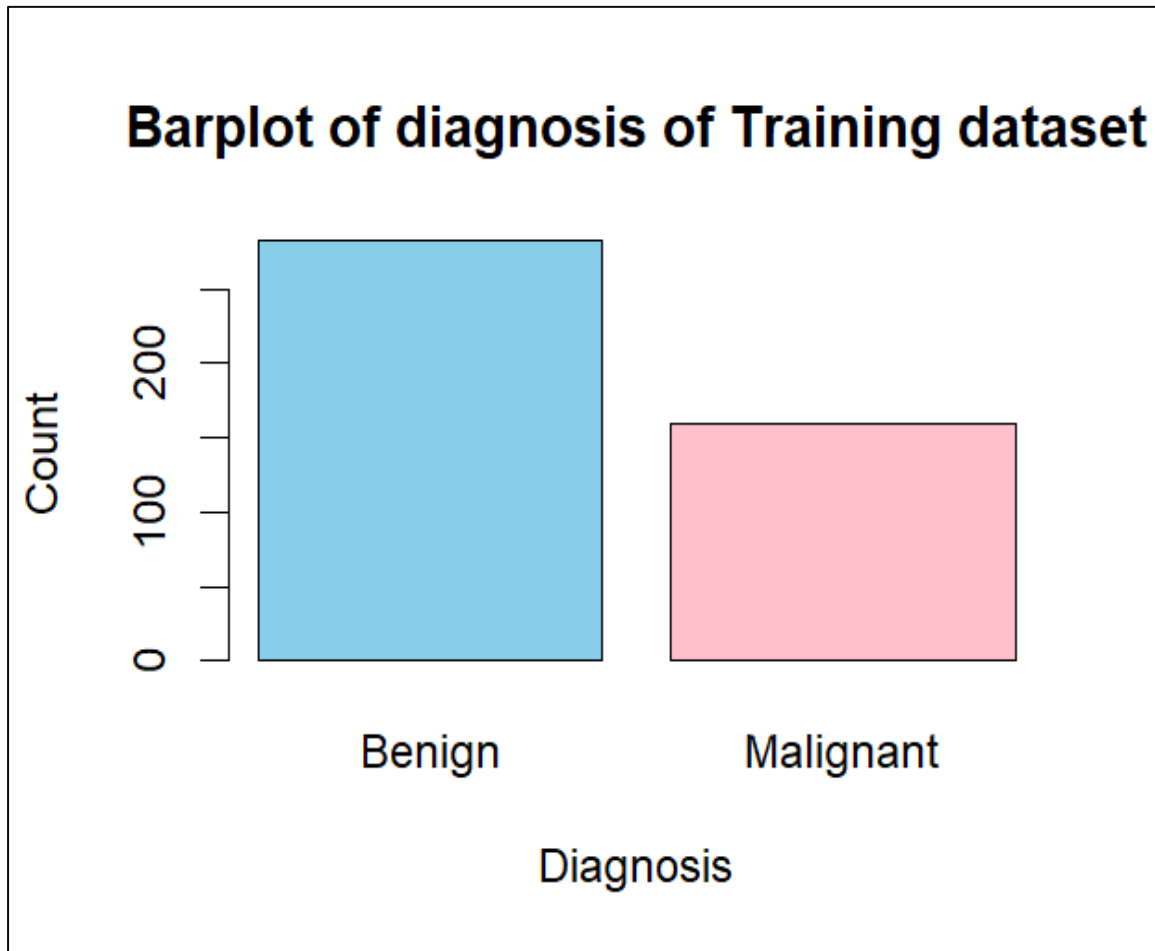
7.4) Splitting the Dataset into *Training* & *Testing* Dataset:

The dataset is divided into two subsets: a training set and a testing set by a ratio of 8:2.

The training set is used to train the model, allowing it to learn patterns and relationships in the data.

The testing set, separate from the training set, is used to evaluate the final model's performance on entirely new, unseen data.

7.4.1 Barplot for the Response Variable of Training Dataset



Here,

total Benign("0") cases are = 296

total Malignant("1")cases are = 159

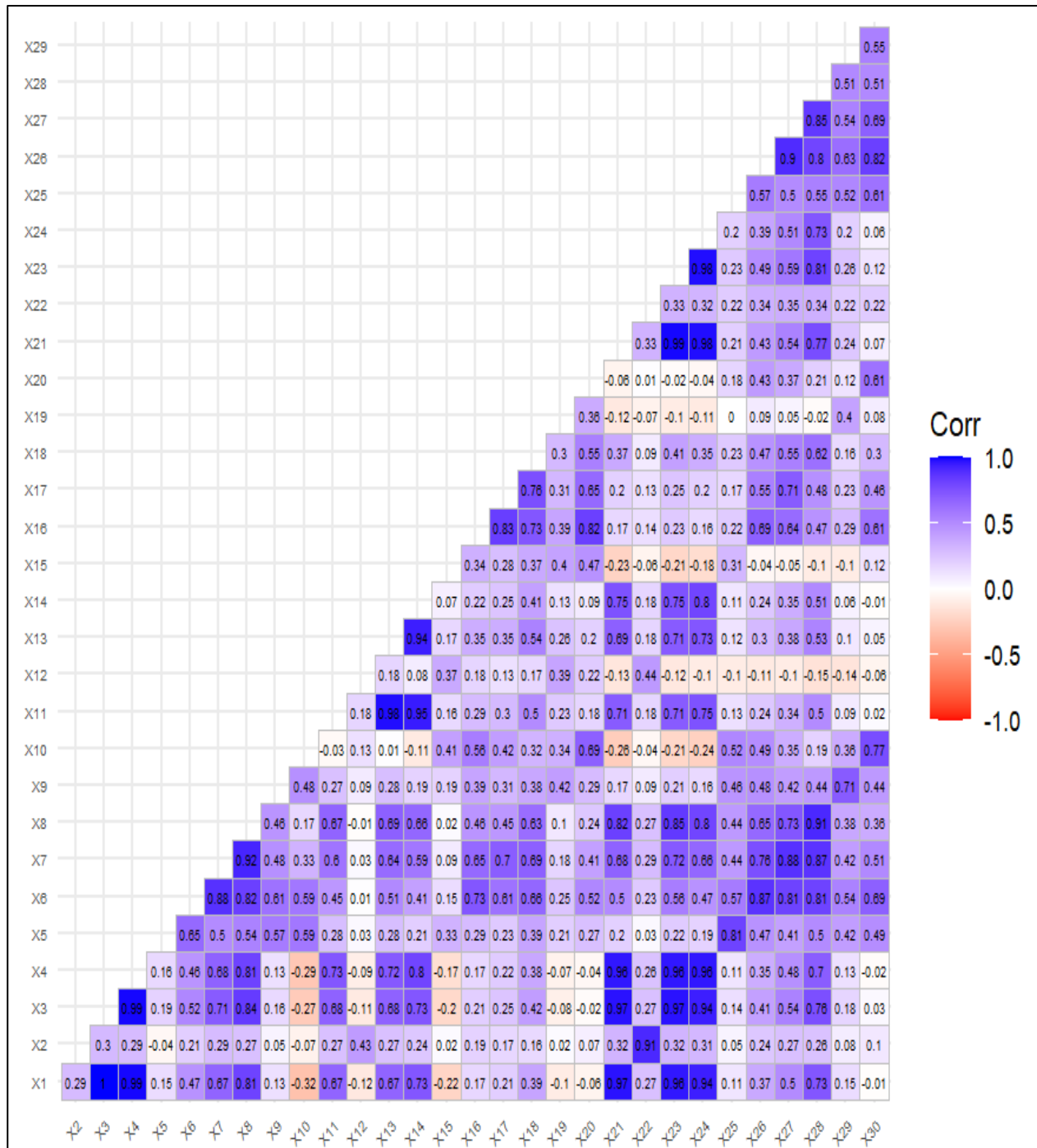
NOTE: We change the names of the variables from diagnosis, radius_mean, texture_mean, perimeter_mean, area_mean,... , fractal_dimension_mean, radius_se,...fractal_dimension_se,..., radius_worst, ..., fractal dimension_worst to "Y", "X1", "X2", "X3", "X4", "X5", "X6", "X7", "X8", "X9", "X10", "X11", "X12", "X13", "X14", "X15", "X16", "X17", "X18", "X19", "X20", "X21", "X22", "X23", "X24", "X25", "X26", "X27", "X28", "X29", "X30" respectively for the sake of simplicity.

7.5) REDUCTION OF PREDICTOR VARIABLES

7.5.1. Scrutiny of Correlation Matrix:

One key assumption about the logistic regression is that the predictor variables should be linearly independent of each other. Through correlation matrix we can observe the correlation of each possible pair of the predictor variables.

The correlation matrix of the 30 variables based on the training dataset is given below-



Comments:➤ **Feature Clustering**

- Features group into three types:

Mean (X1 to X10): average values.

Se (X11 to X20): standard error (variation).

Worst (X21 to X30): largest (most abnormal) values.

These clusters are more **internally correlated**.

- This image is a correlation heatmap showing the pairwise Pearson correlation coefficients between all features in the Wisconsin Breast Cancer Dataset. The values range from -1 to 1, where:
 - 1 = perfect positive correlation
 - -1 = perfect negative correlation
 - 0 = no correlation

 Key Takeaways

(1) i) *Radius mean(X1)*, *Perimeter mean(X3)*, and *Area mean(X4)* are **strongly correlated** (close to 0.99), implying redundancy.

ii) *Radius se(X11)*, *Perimeter se(X13)*, *Area se(X14)* are **highly correlated** within themselves; the correlations being ≥ 0.93 .

iii) *Radius worst(X21)*, *Perimeter worst(X23)*, and *Area worst(X24)* also show **very high correlation**, also mirroring the "mean" values (correlation being >0.98)

(2) i) *Compactness mean(X6)*, *Concavity mean(X7)*, *Concave-points mean(X8)* are **highly correlated** within themselves; the correlations being ≥ 0.83 .

ii) Similarly, *Compactness se(X16)*, *Concavity se(X17)*, *Concave-points se(X18)* are **highly correlated** within themselves; the correlations being ≥ 0.73 .

iii) Also, *Compactness worst(X26)*, *Concavity worst(X27)*, *Concave-points worst(X28)* are **highly correlated** within themselves; the correlations being ≥ 0.8 .

(3) The variables giving the *mean measurements* are **highly correlated** with the corresponding variables giving *worst measurements*; the correlations ranging from 0.71 to 0.97 (rightly so, as higher values of a variable push the mean of the variable higher).

7.5.2) Examining the pairwise scatter plot (Scatter-Plot Matrix):

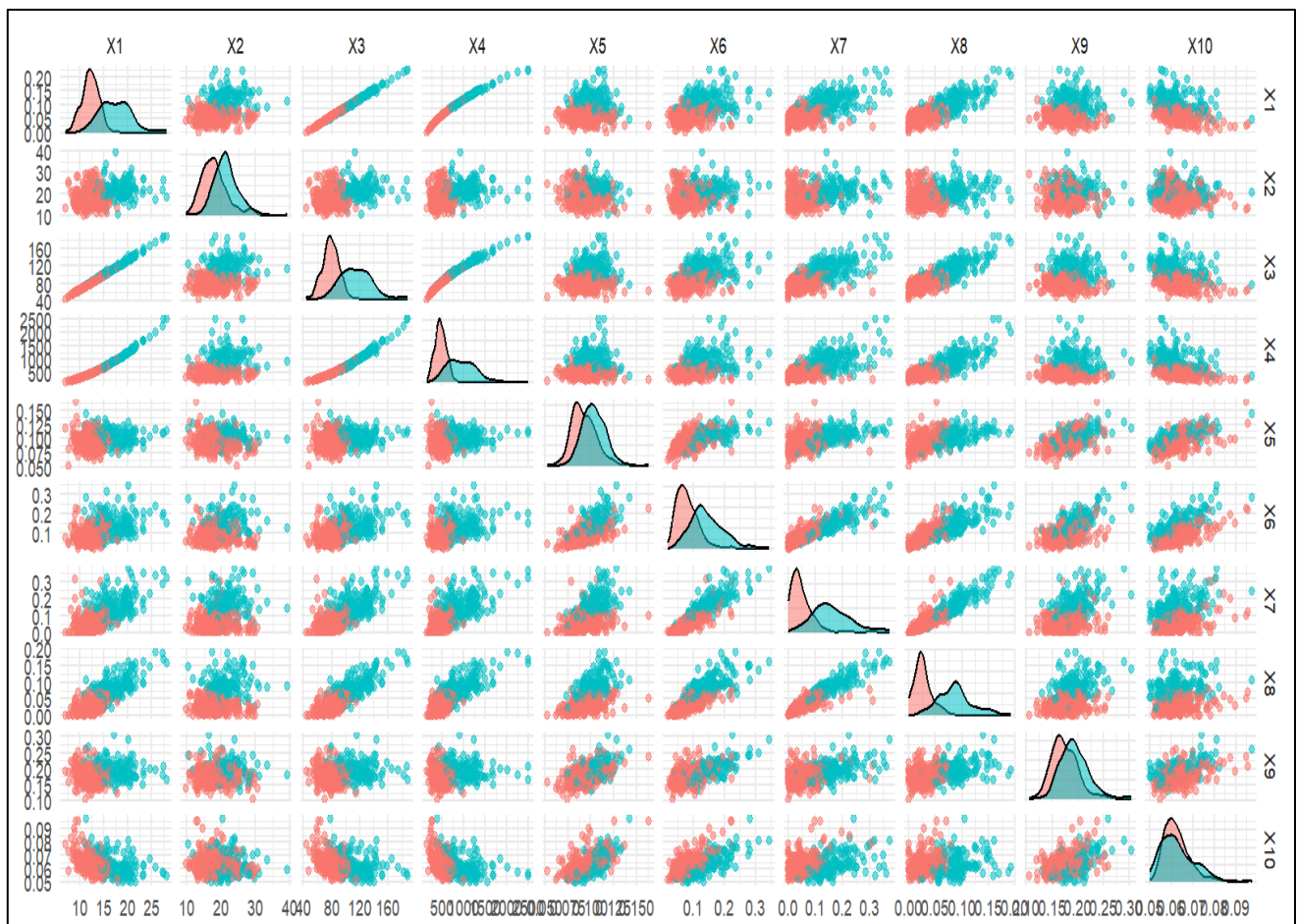
The following figure presents a **pairplot** combining **scatterplots** and **kernel density estimates (KDEs)** for the 10 mean features of the Breast Cancer Dataset. Each data point is color-coded by diagnosis: **pink** for benign and **skyblue** for malignant tumors.

On the **diagonal**, KDE plots visualize the **distribution of each feature** separately for benign and malignant cases. Features with **minimal overlap between the two KDE** indicate **strong discriminatory power**, making them valuable predictors for breast cancer classification.

The **off-diagonal scatterplots** show pairwise relationships between features. Strong linear relationships highlight potential **multicollinearity**, which should be considered during model building and feature selection.

This visualization helps identify:

- Which features best separate benign from malignant cases.
- Redundant features with strong linear associations.
- Weak predictors with overlapping class distributions



Comments: 1. Clear KDE Separation = Strong Predictors

- **Radius mean, perimeter mean, and area mean** show **well-separated KDE peaks**:
 - Benign tumors cluster at **lower values**
 - Malignant tumors shift to **higher values**
 - This confirms these are **very strong predictors** of malignancy.

 2. Moderate KDE Overlap

- Features like **concavity mean, concave-points mean, and compactness mean** still show **some separation**, but with **slightly more overlap**.
 - These features still help predict malignancy, though **less distinct** than size-based features.

 3. Weak KDE Separation = Poor Predictors

- **Smoothness mean, symmetry mean, and especially fractal dimension mean** have **heavily overlapping KDEs**, meaning:
 - The feature values for benign and malignant are **not clearly different**
 - These are **weak standalone predictors**

 4. Scatterplots Confirm Feature Relationships

- Features like radius mean, perimeter mean, and area mean show **tight linear relationships**
- Suggests possible **redundancy**—one might be chosen among these 3 features (e.g., radius mean) to avoid multicollinearity.

 Key Takeaways

- (1) There is an almost **perfect linear relationship** between *radius mean* and *perimeter mean*.
- (2) A **non-linear relation** between *radius mean* and *area mean* is evident.
- (3) *perimeter mean* and *area mean* also show a **non-linear relationship**.
- (4) Scatterplot of *concavity mean* and *compactness mean* shows a **linear relationship** between them.
- (5) A quick go-through of the KDEs available along the principle diagonal of the scatterplot matrix reveals that except for *fractal dimension mean*, KDEs of Malignant are always shifted to the right of KDEs of Benign; implying larger values of the variables tend to malignancy.

7.5.3. Variable reduction- Phase I

1. We have seen that *radius mean*(X1), *perimeter mean*(X3) and *area mean*(X4) are **highly correlated** among themselves. If we assume the shape of a nucleus roughly as a circle then radius mean stands to be the most elementary measurement of shape of the nucleus. Also radius mean is directly interpretable and simpler. So we keep *radius mean*(X1) and drop the other two from our data-set.

[Supporting source: "Analysis of 569 fine-needle aspirate samples showed that radius_mean alone generated a strong separation between benign ($\approx 12.1 \mu\text{m}$) and malignant tumors ($\approx 17.5 \mu\text{m}$), supporting its use as a standalone diagnostic feature in breast cancer modeling" (Author, 2023)]

https://www.researchgate.net/publication/367378275_Analysis_of_Impact_of_Principal_Component_Analysis_and_Feature_Selection_for_Detection_of_Breast_Cancer_Using_Machine_Learning_Algorithms

And for the same reason we keep *radius se*(X11) and *radius worst*(X21) and drop *perimeter se*(X13), *perimeter worst*(X23) and *area se*(X14), *area worst*(X24).

2. *Compactness mean*(X6), *concave points mean*(X8), *compactness se*(X16), *concave points se*(X18), *compactness worst*(X26), *concave points worst*(X28) are also dropped.

Also the Testing dataset should be adjusted accordingly.

7.5.4. Linearity assumptions checking

Another key assumption of Logistic regression is linear relationship between the log odds and the continuous predictor variables.

We can check linearity in the following way:

- I. We divide the observed values of predictor into 4 subsets individually. The splitting should be based on their quartiles.

Let X be the numeric value of a predictor variable.

- Subset 1 consists those X values which are $< Q1$.
- Subset 2 consists those X values which are $> Q1$ and $< Q2$.
- Subset 3 consists those X values which are $> Q2$ and $< Q3$.
- Subset 4 consists those X values which are $> Q3$.

Here Q1, Q2, Q3 are 1st, 2nd and 3rd quartile respectively.

- II. Now we need to calculate the number of Malignant (1) cases for each subset. And then calculate the estimated probability of having malignancy.

$$p_i = \frac{\text{Total number of Malignant cases in subset } i}{\text{Total number of observed value in subset } i}, \quad \forall i = 1(1) 4$$

- III. The log-odds are given by, $l_i = \ln \left(\frac{p_i}{1-p_i} \right), \quad \forall i = 1(1) 4$

Let, the log-odds are l_1, l_2, l_3, l_4 .

- IV. Obtain the medians of the 4 subsets. Let m_1, m_2, m_3, m_4 be the medians of the subsets respectively.
- V. Plot the $\text{logit}(\text{log-odds})$ of the outcome against these medians. The points are $(m_1, l_1), (m_2, l_2), (m_3, l_3)$ and (m_4, l_4) . Then join the four points by line segments.

Interpretations:

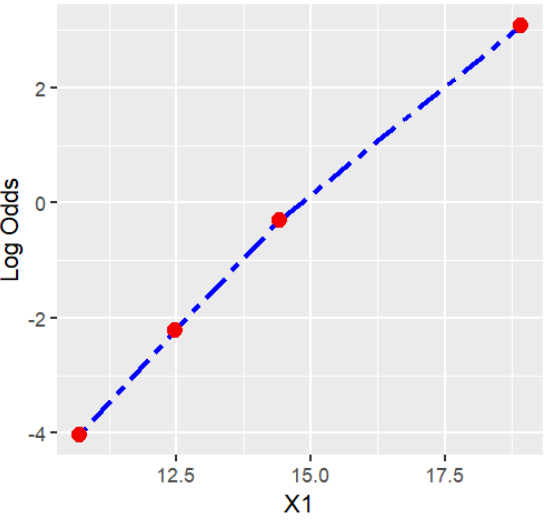
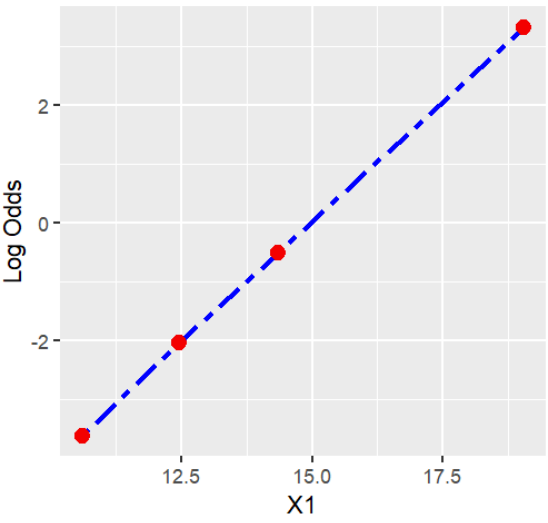
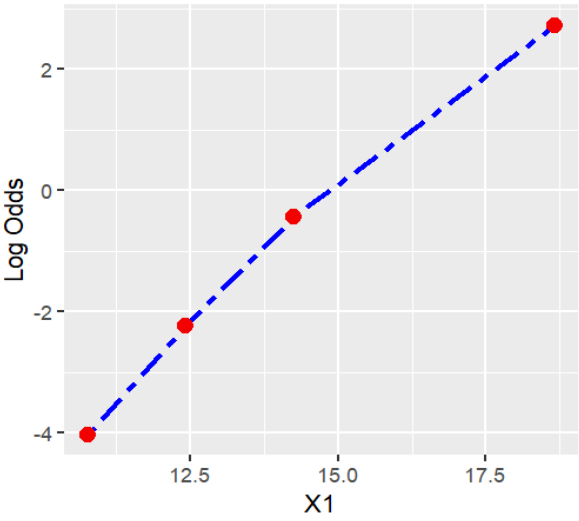
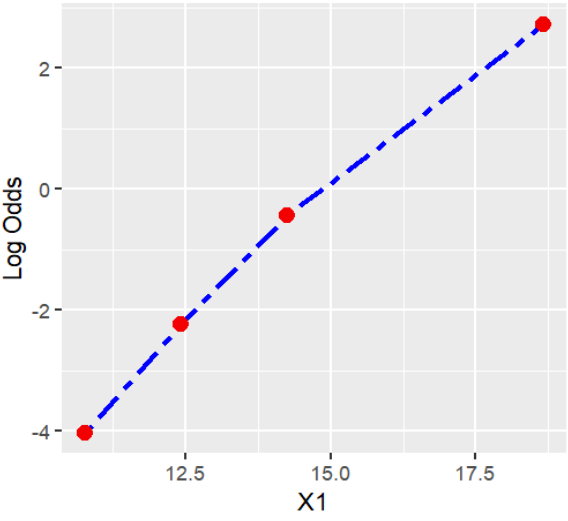
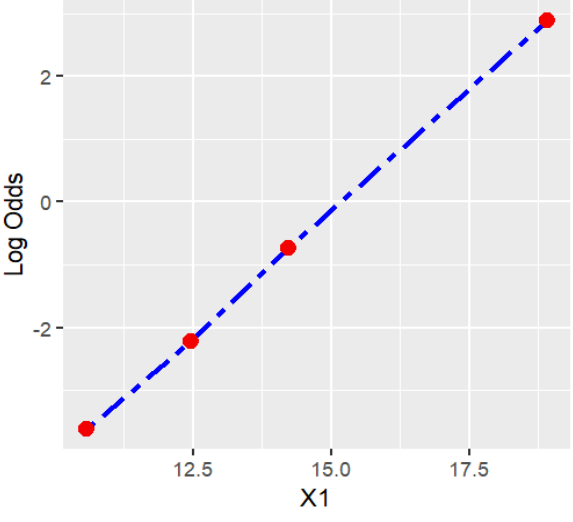
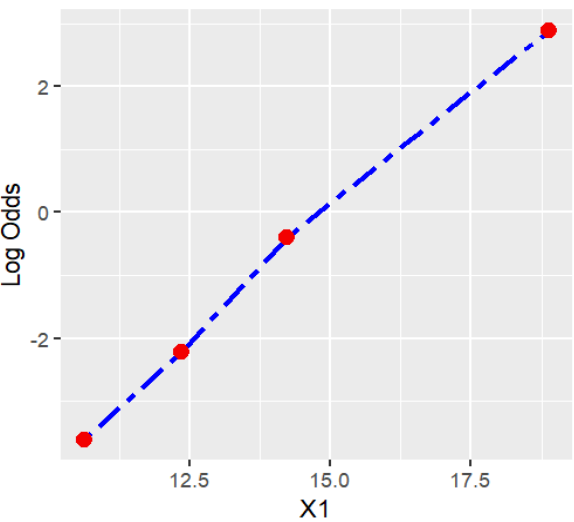
- If the points form a straight or near linear line, the linearity assumption holds.
- If the line is curved, S- shaped or irregular, then this suggests non- linearity in the predictor's relationship with the logit. In this case we will drop those predictor variables.

Doing the linearity check on multiple training sets will help us to draw more robust and reliable conclusions. So, we create 5 more training sets and repeat the same for every sets.

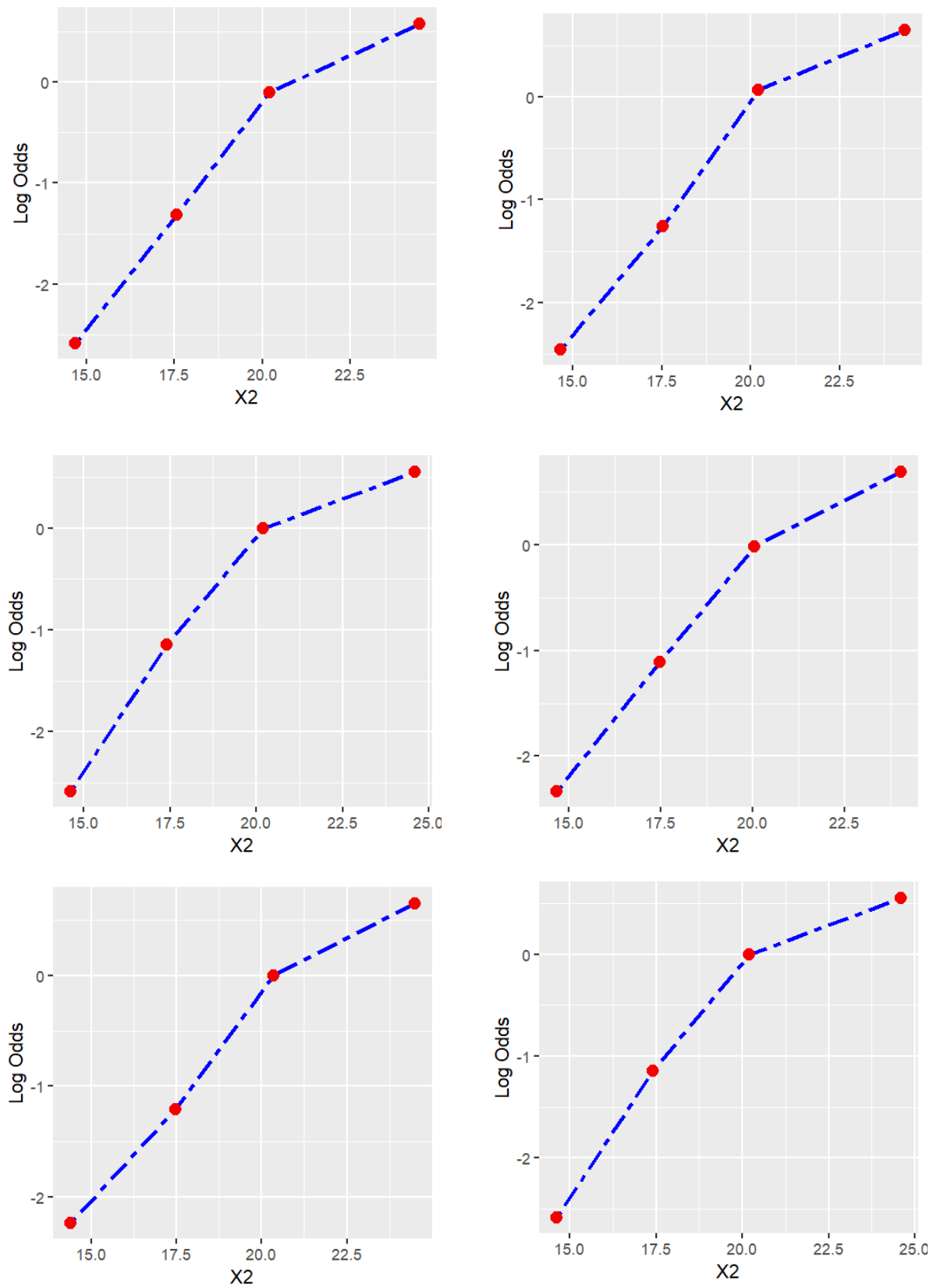
After the phase-I reduction we have total 13 variables remaining. These are –

1. Radius mean(X1).
2. Texture mean(X2).
3. Smoothness mean(X5).
4. Concavity mean(X7).
5. Symmetry mean(X9).
6. Fractal dimension mean(X10).
7. Radius se(X11).
8. Texture se(X12).
9. Smoothness se(X15).
10. Concavity se(X17)
11. Symmetry se(X19)
12. Fractal dimension se(X20)
13. Radius worst(X21).
14. Texture worst(X22).
15. Smoothness worst(X25).
16. Concavity worst(X27).
17. Symmetry worst(X29).
18. Fractal dimension worst(X30)

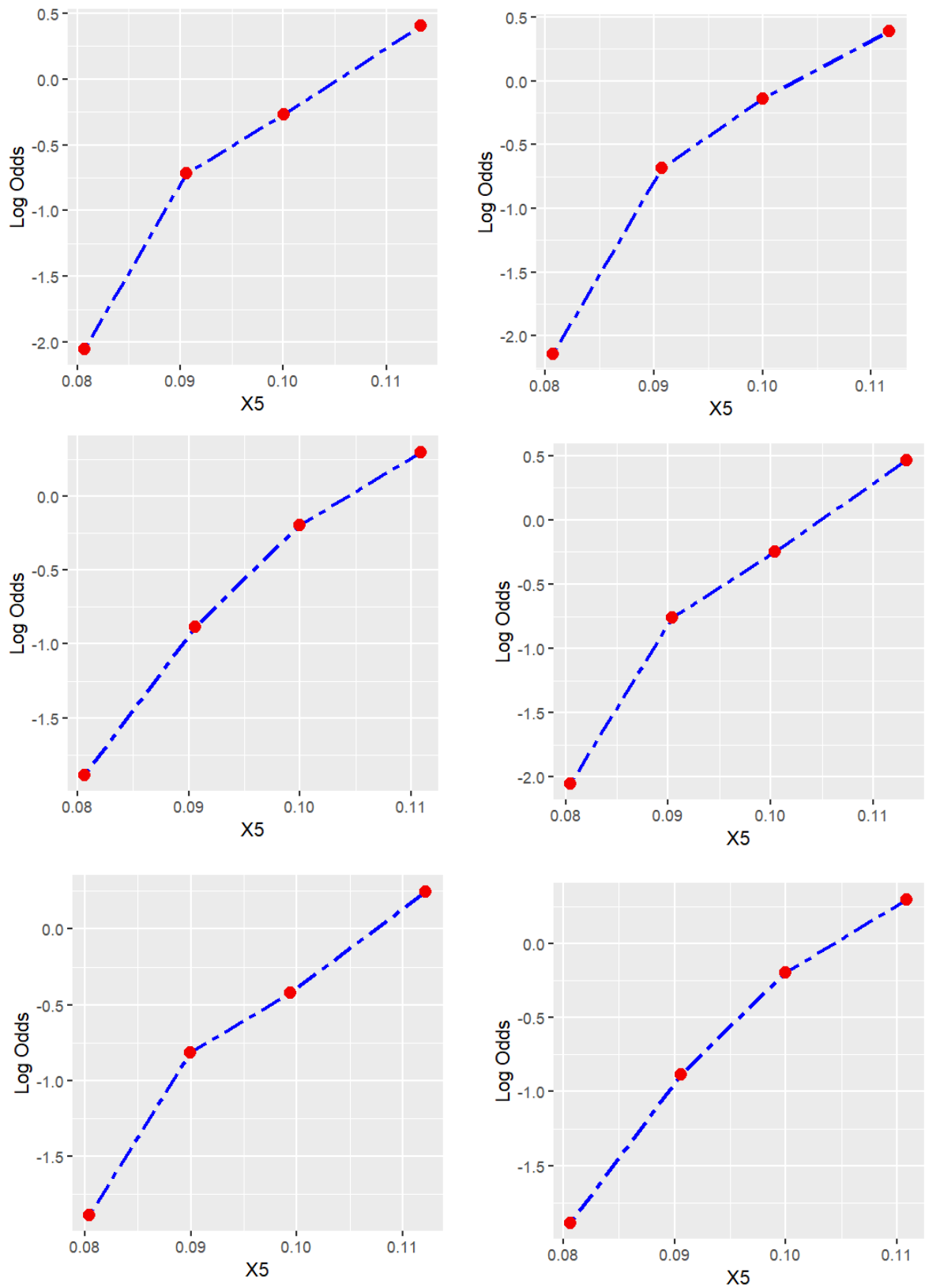
LINEARITY CHECK FOR RADIUS MEAN(X1)



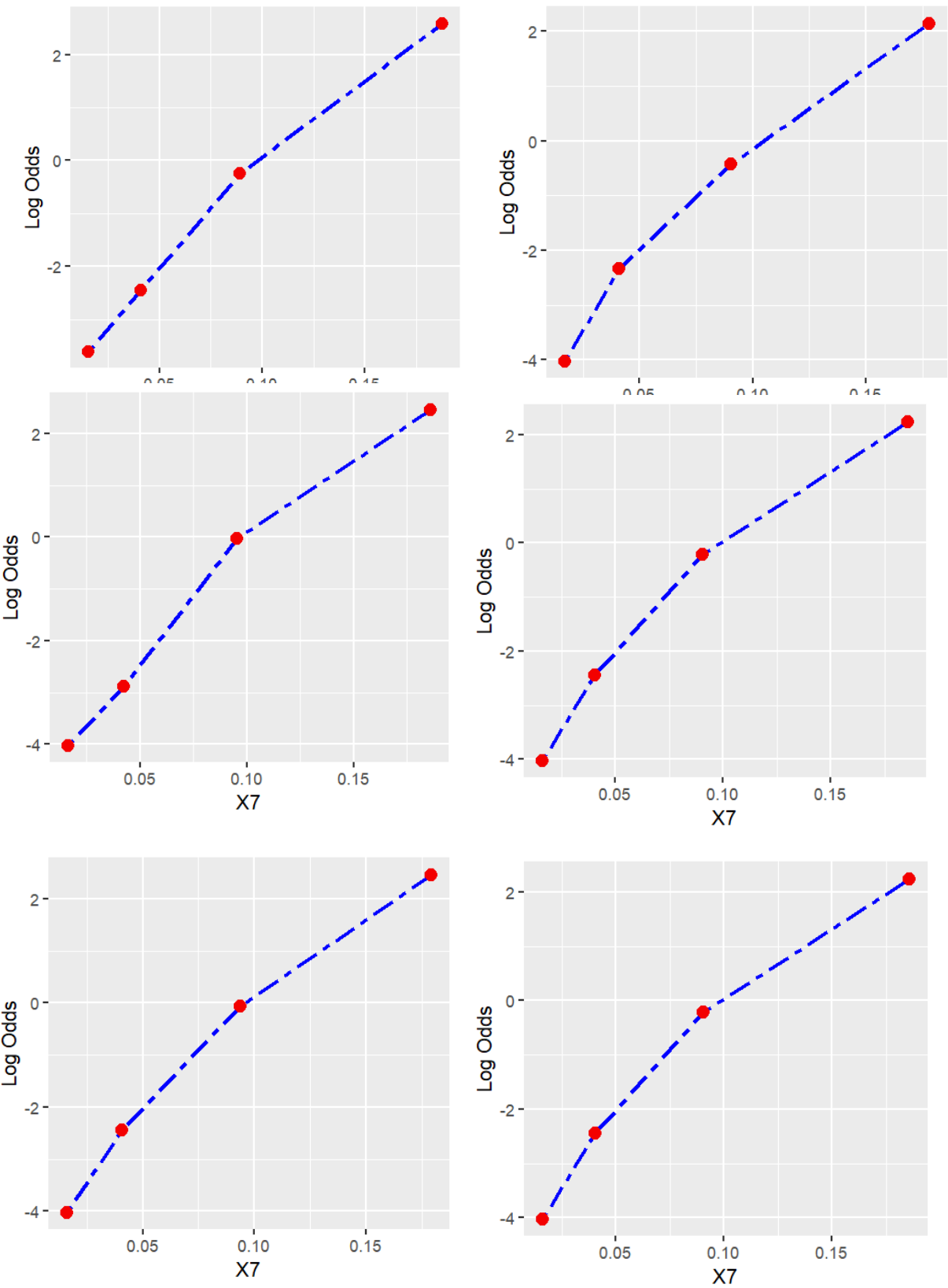
LINEARITY CHECK FOR TEXTURE MEAN(X2)



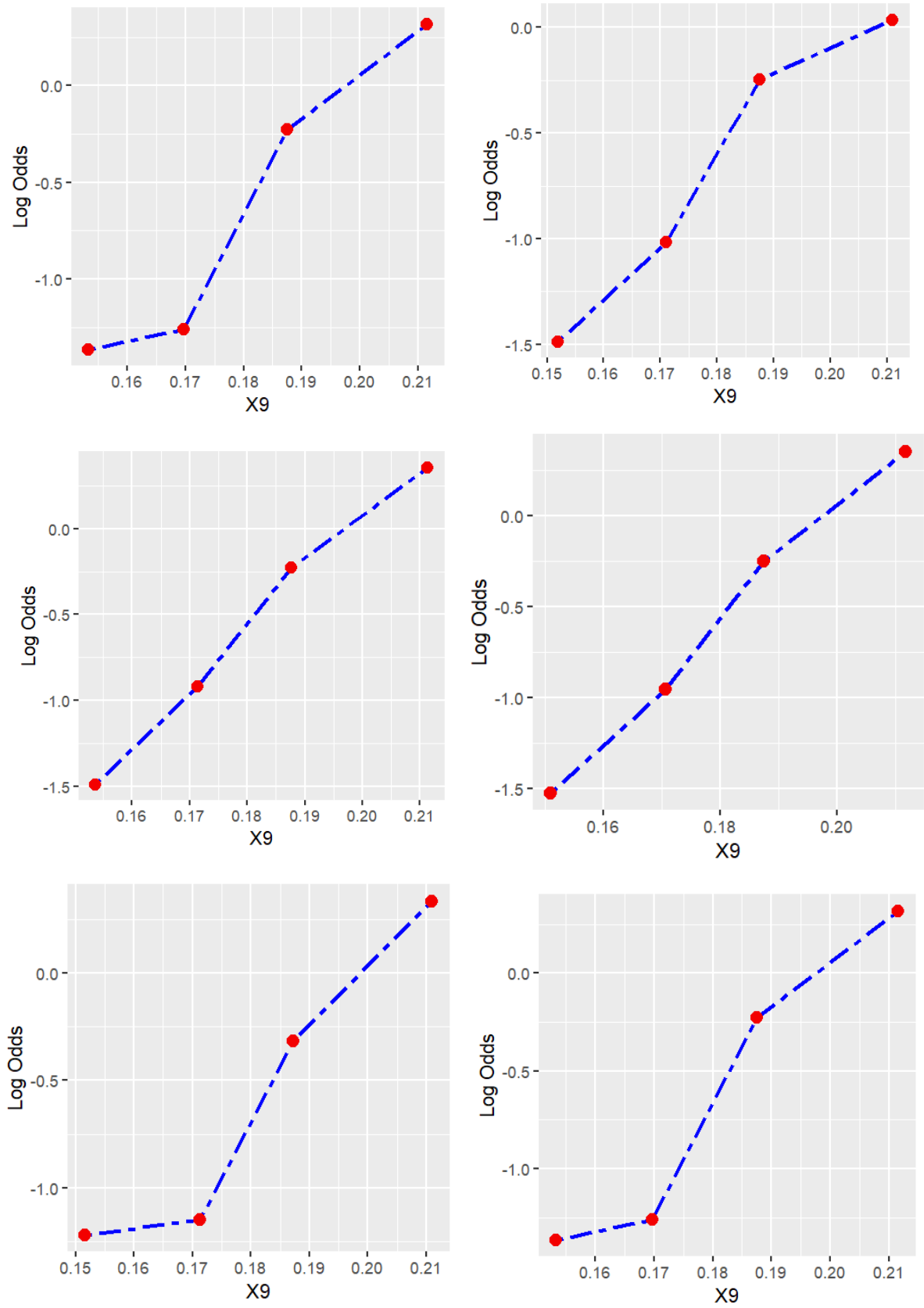
LINEARITY CHECK FOR SMOOTHNESS MEAN(X5)



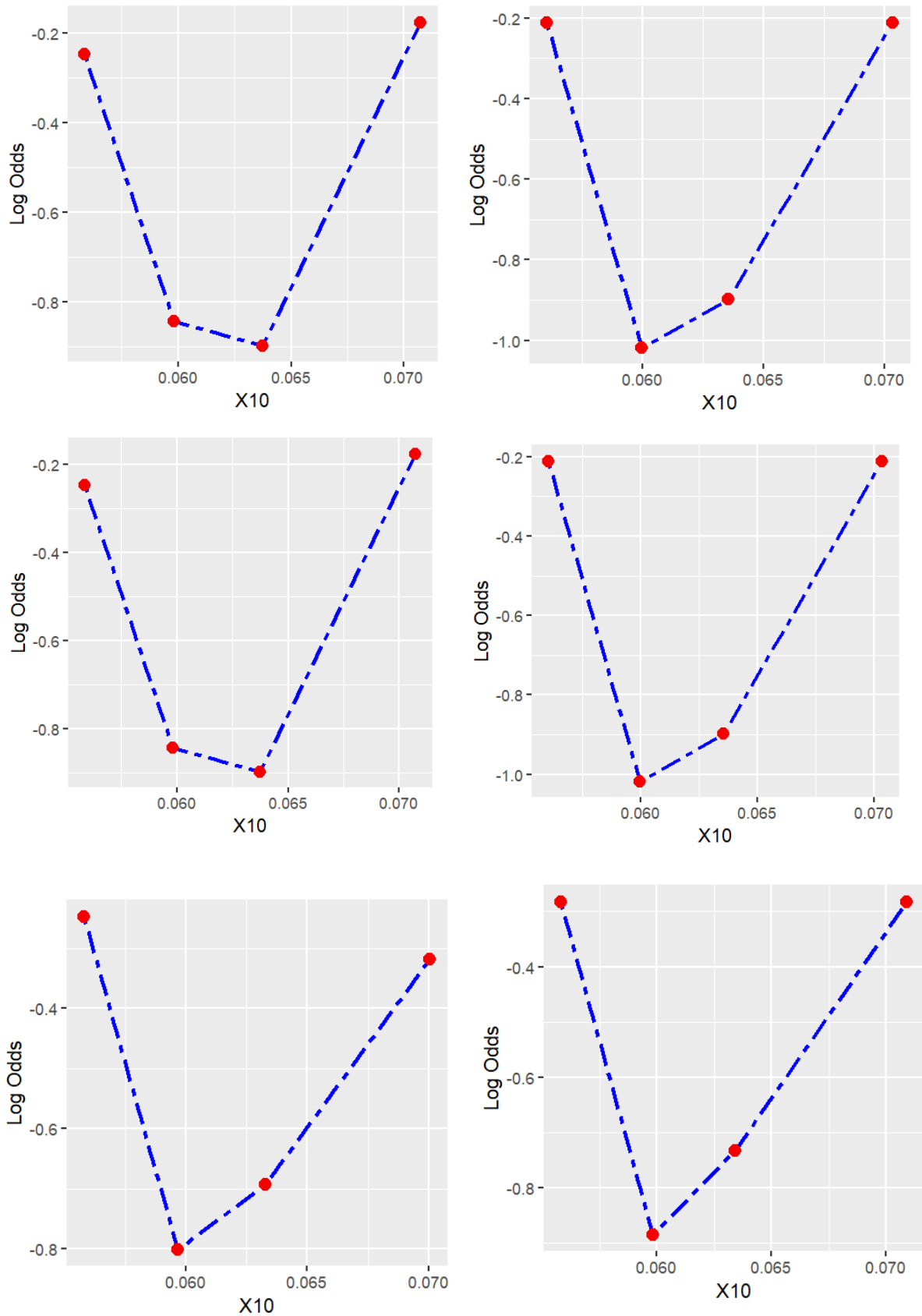
LINEARITY CHECK FOR CONCAVITY MEAN(X7)



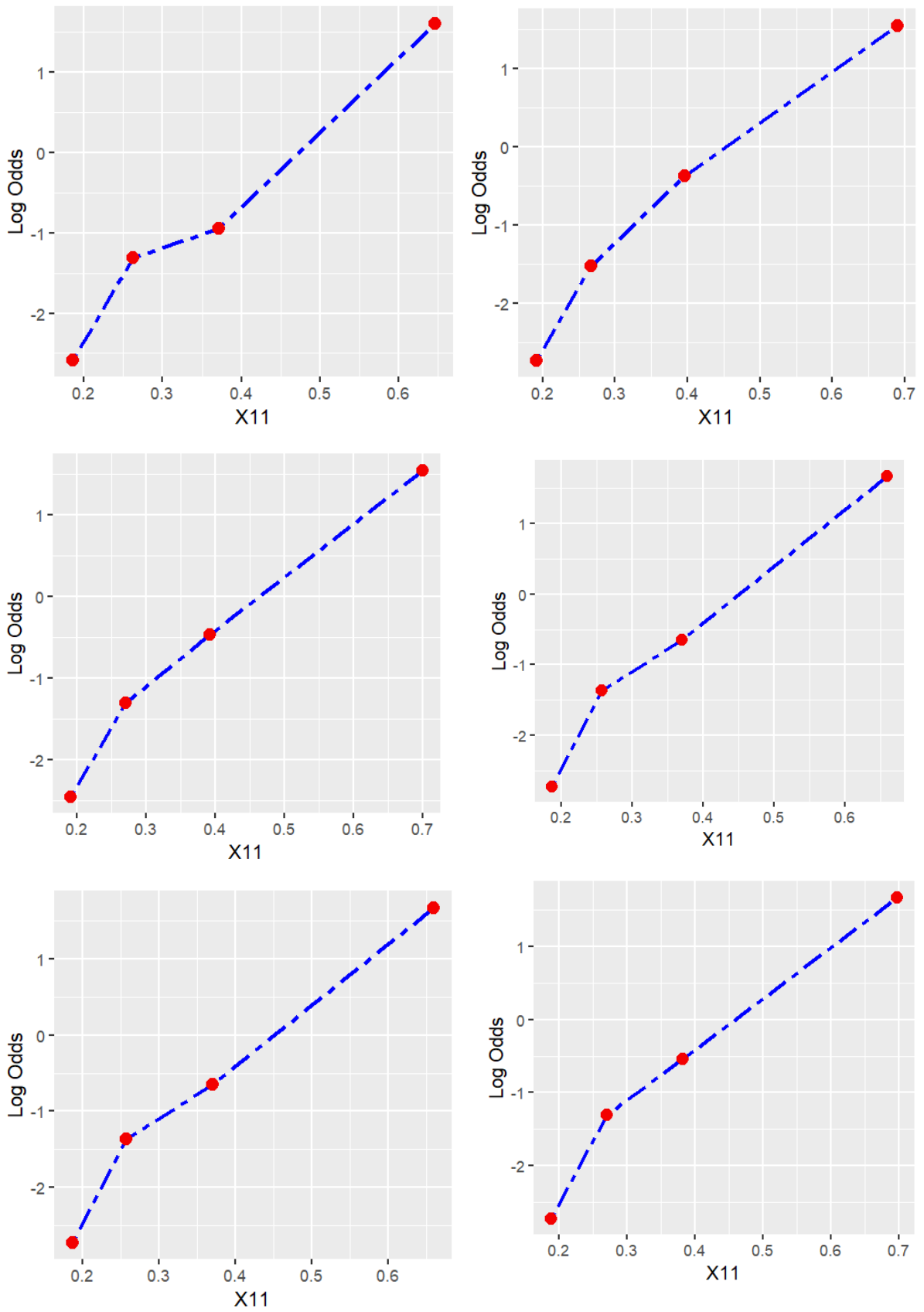
LINEARITY CHECK FOR SYMMETRY MEAN(X9)



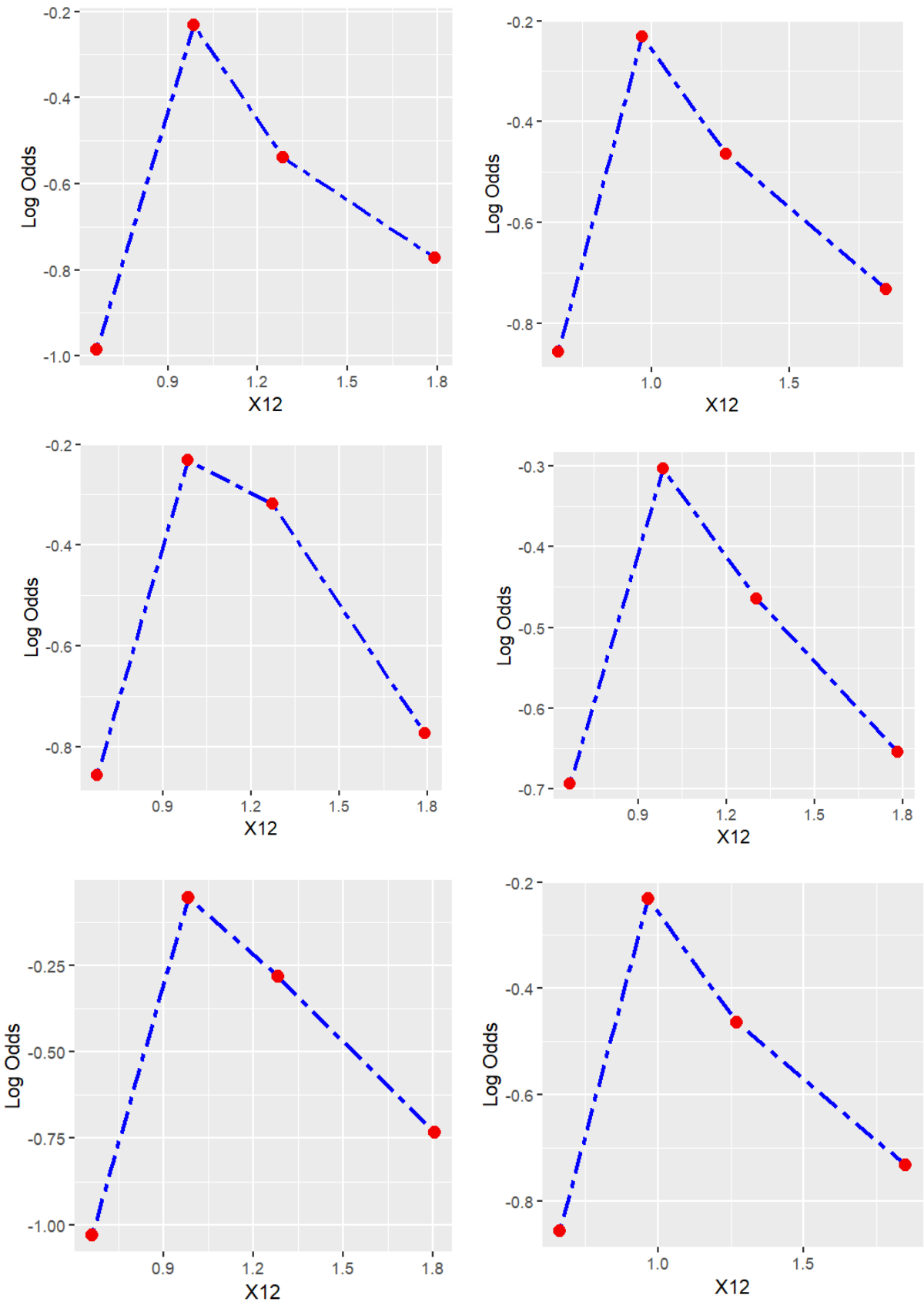
LINEARITY CHECK FOR FRACTAL DIMENSION MEAN(X10)



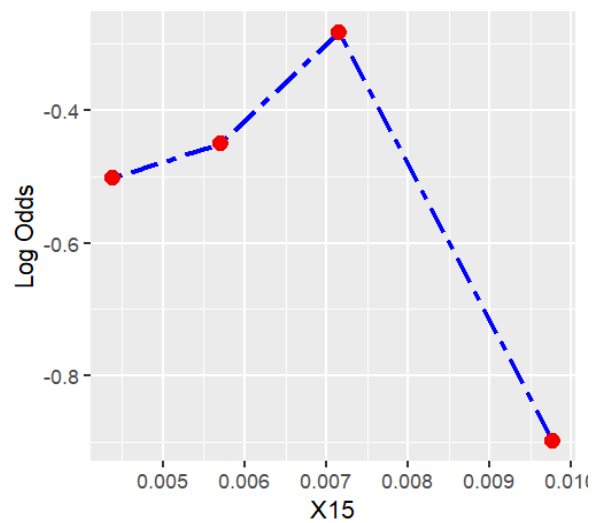
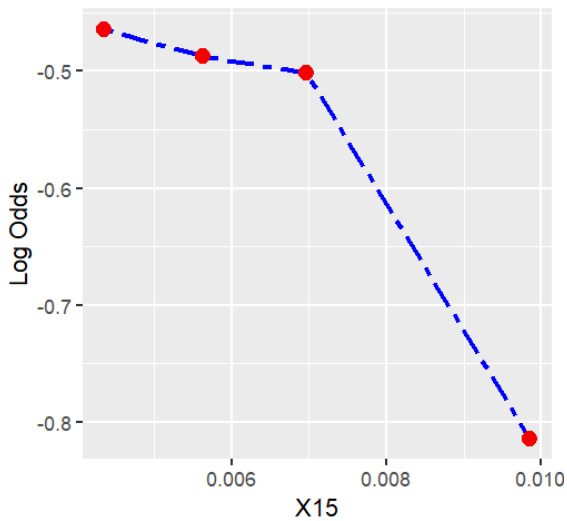
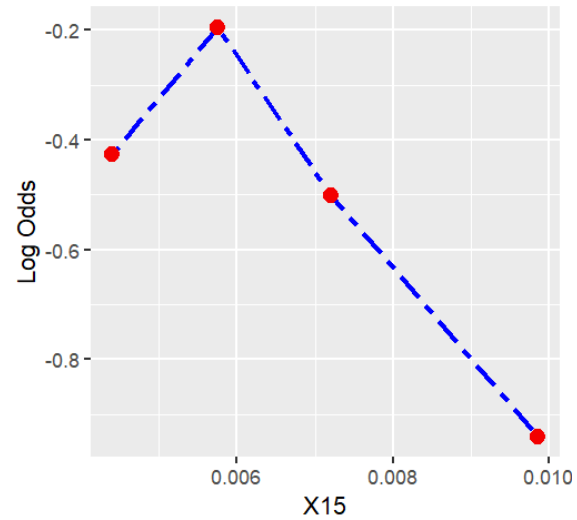
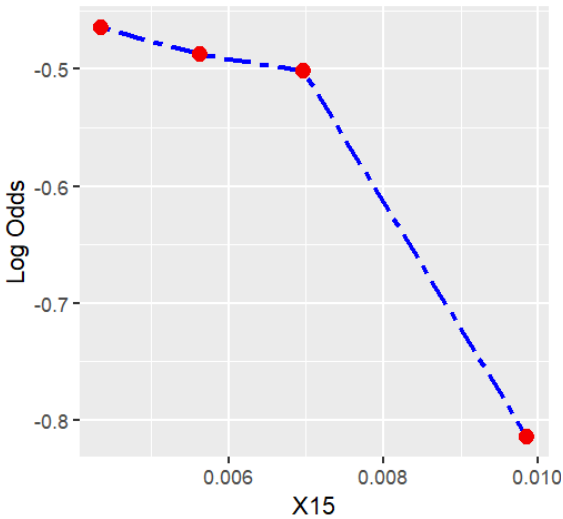
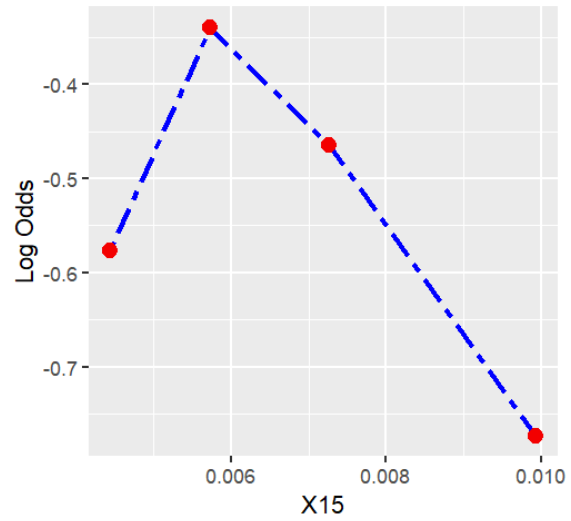
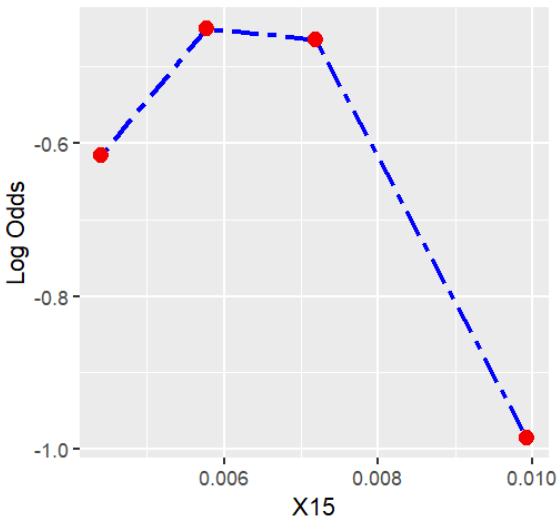
LINEARITY CHECK FOR RADIUS SE(X11)



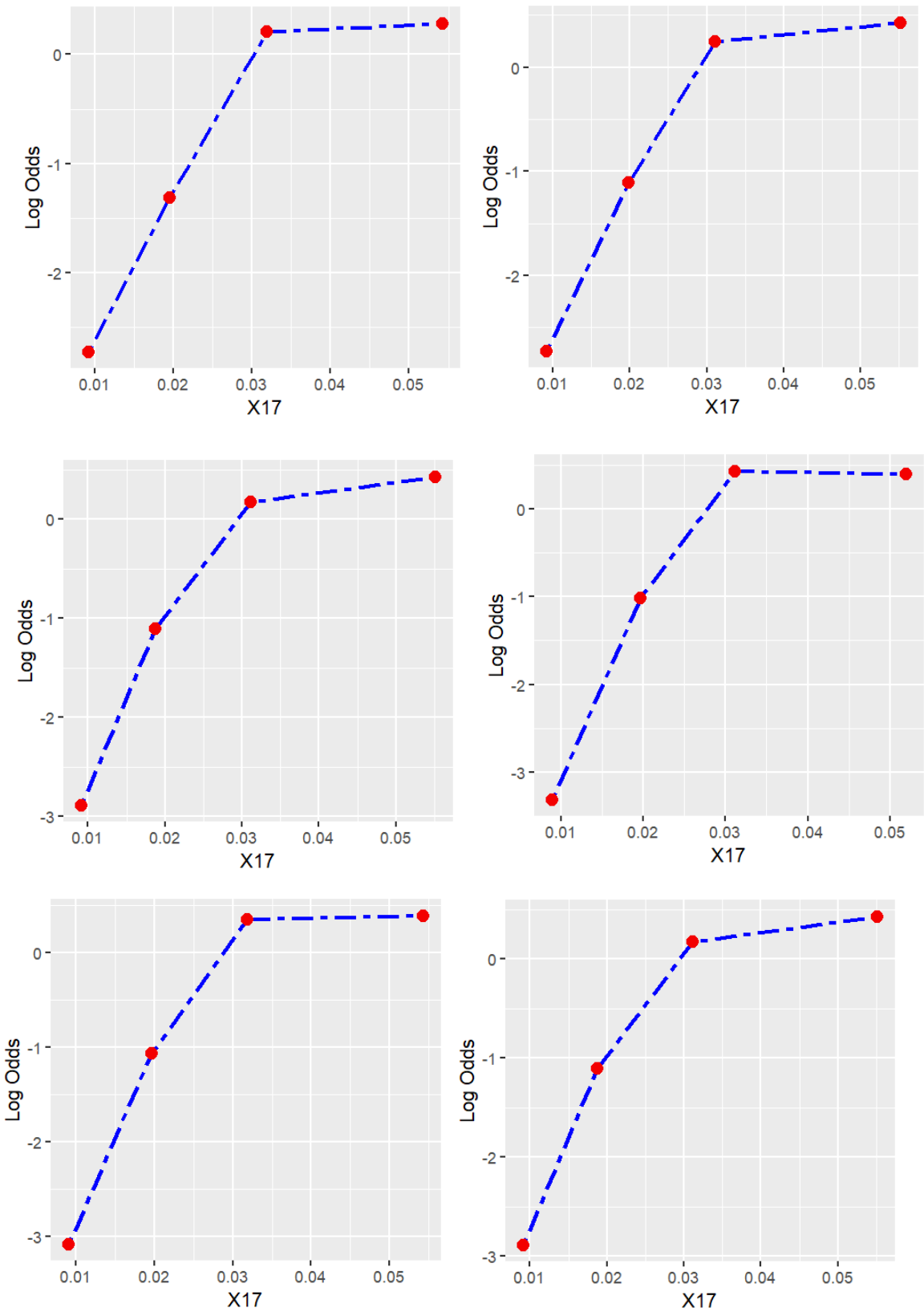
LINEARITY CHECK FOR TEXTURE SE(X12)



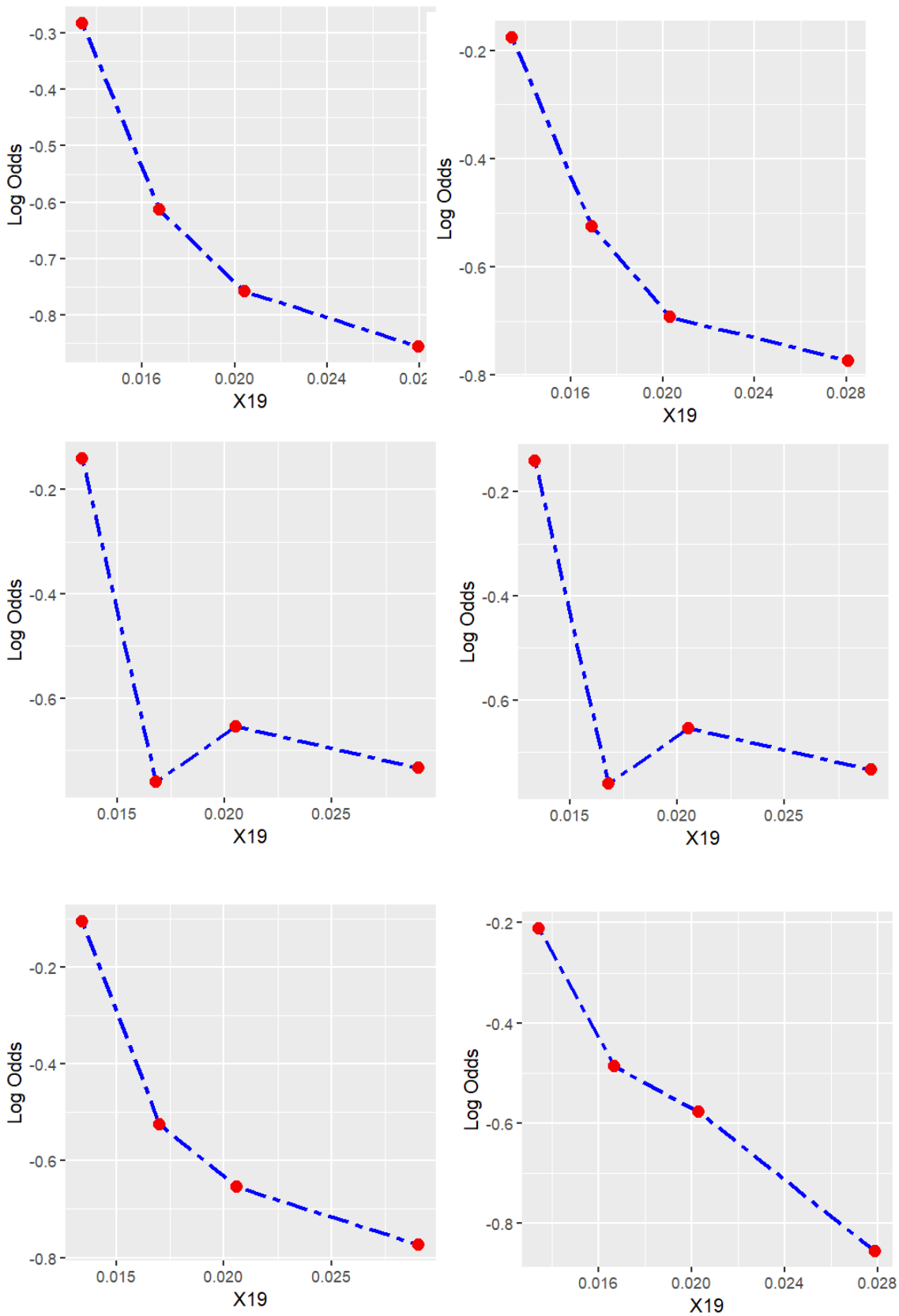
LINEARITY CHECK FOR SMOOTHNESS SE(X15)



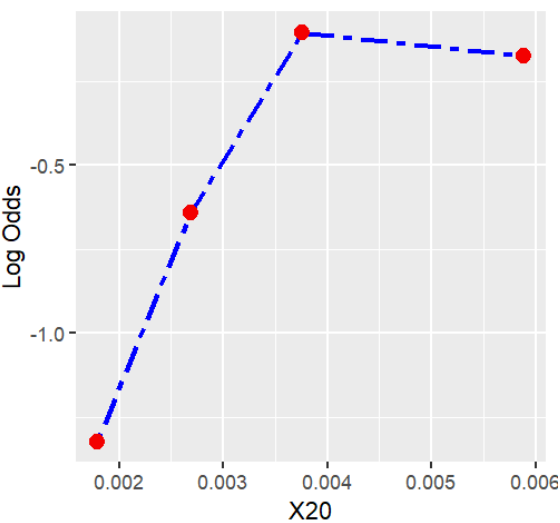
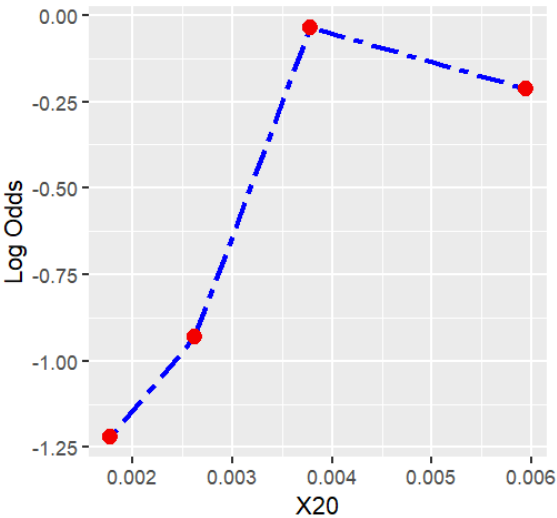
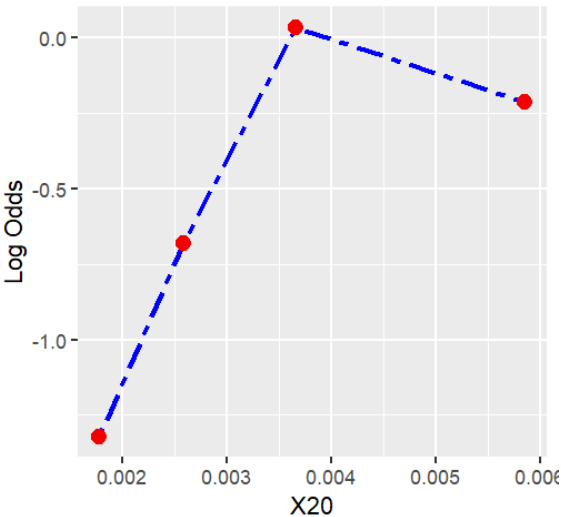
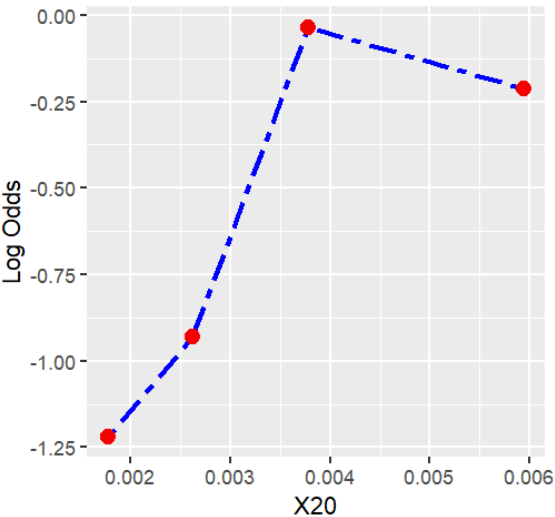
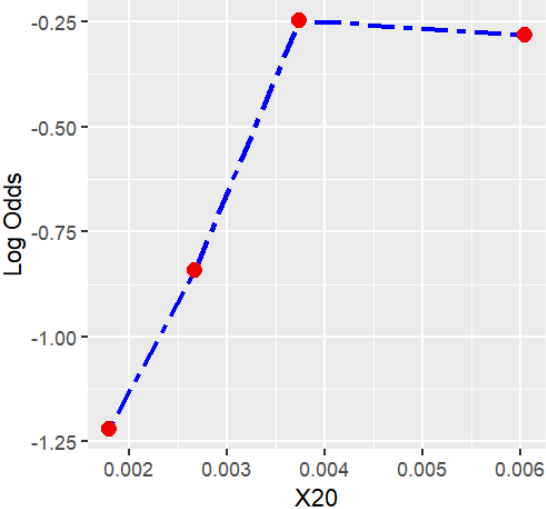
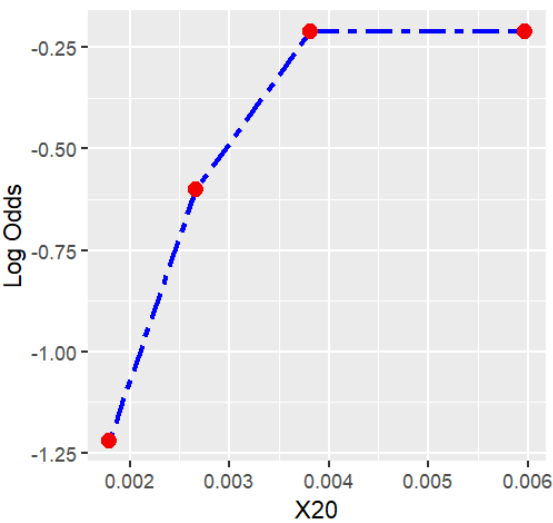
LINEARITY CHECK FOR CONCAVITY SE(X17)



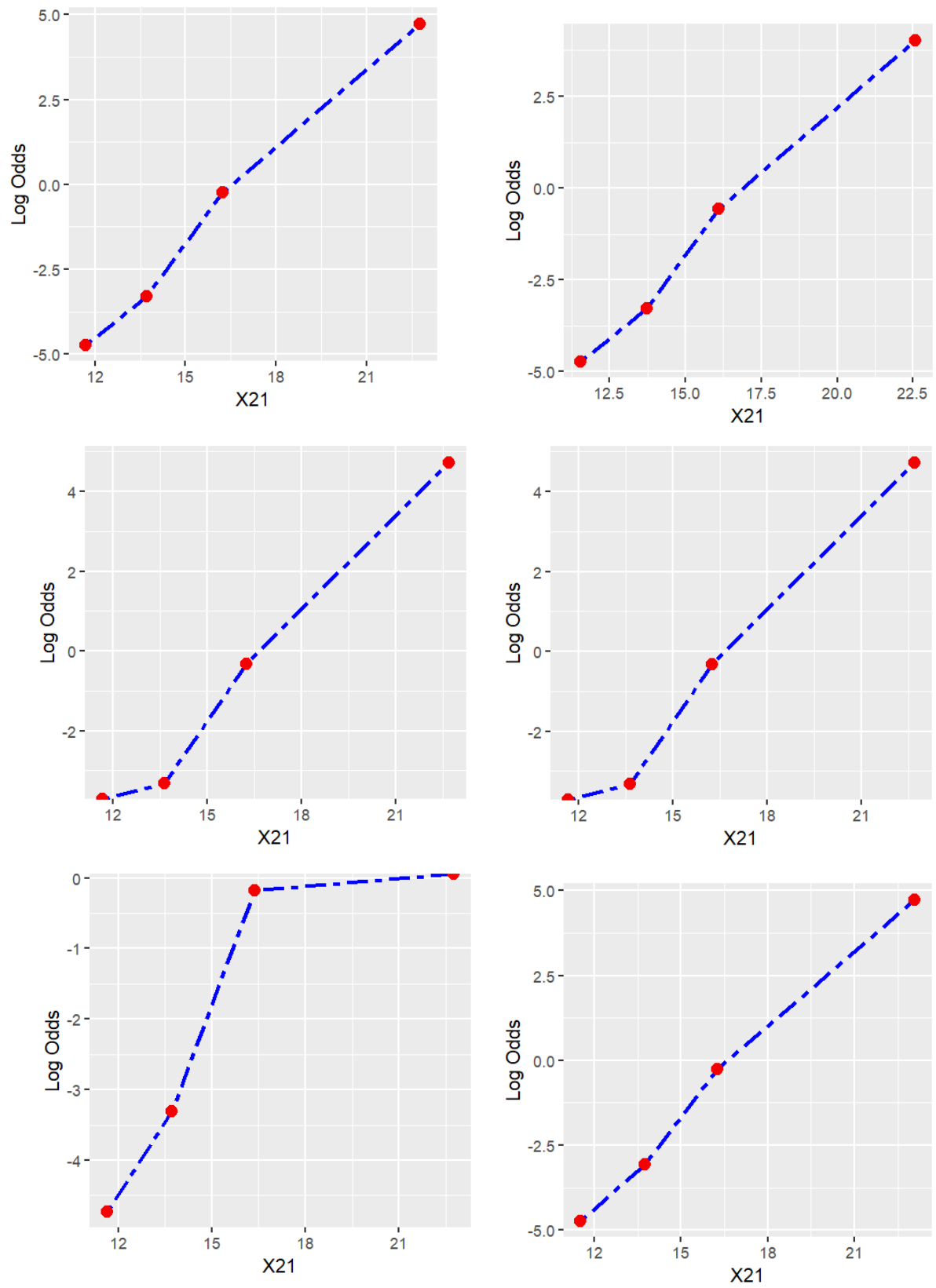
LINEARITY CHECK FOR SYMMETRY SE(X19)



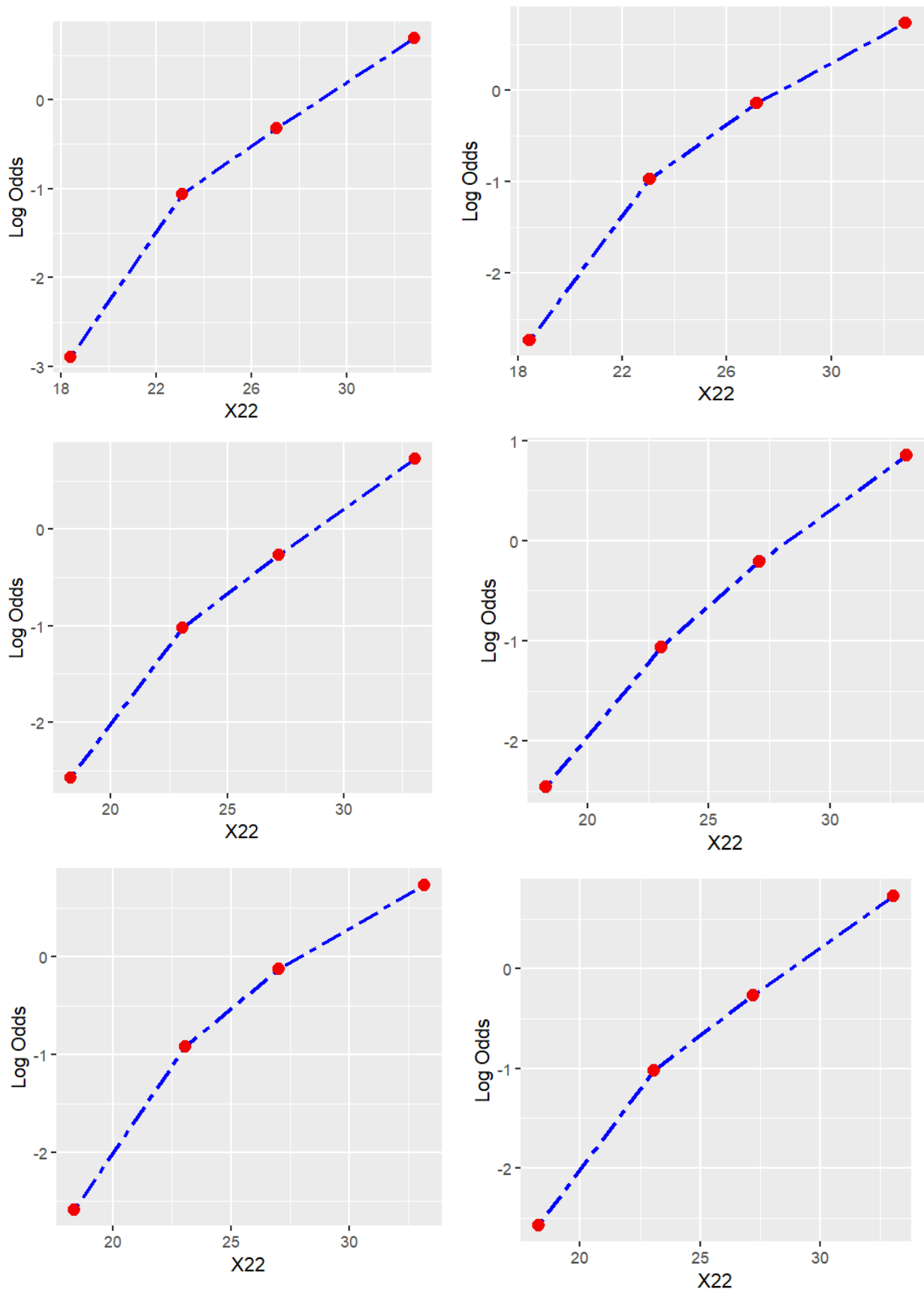
LINEARITY CHECK FOR FRACTAL DIMENSION SE(X20)



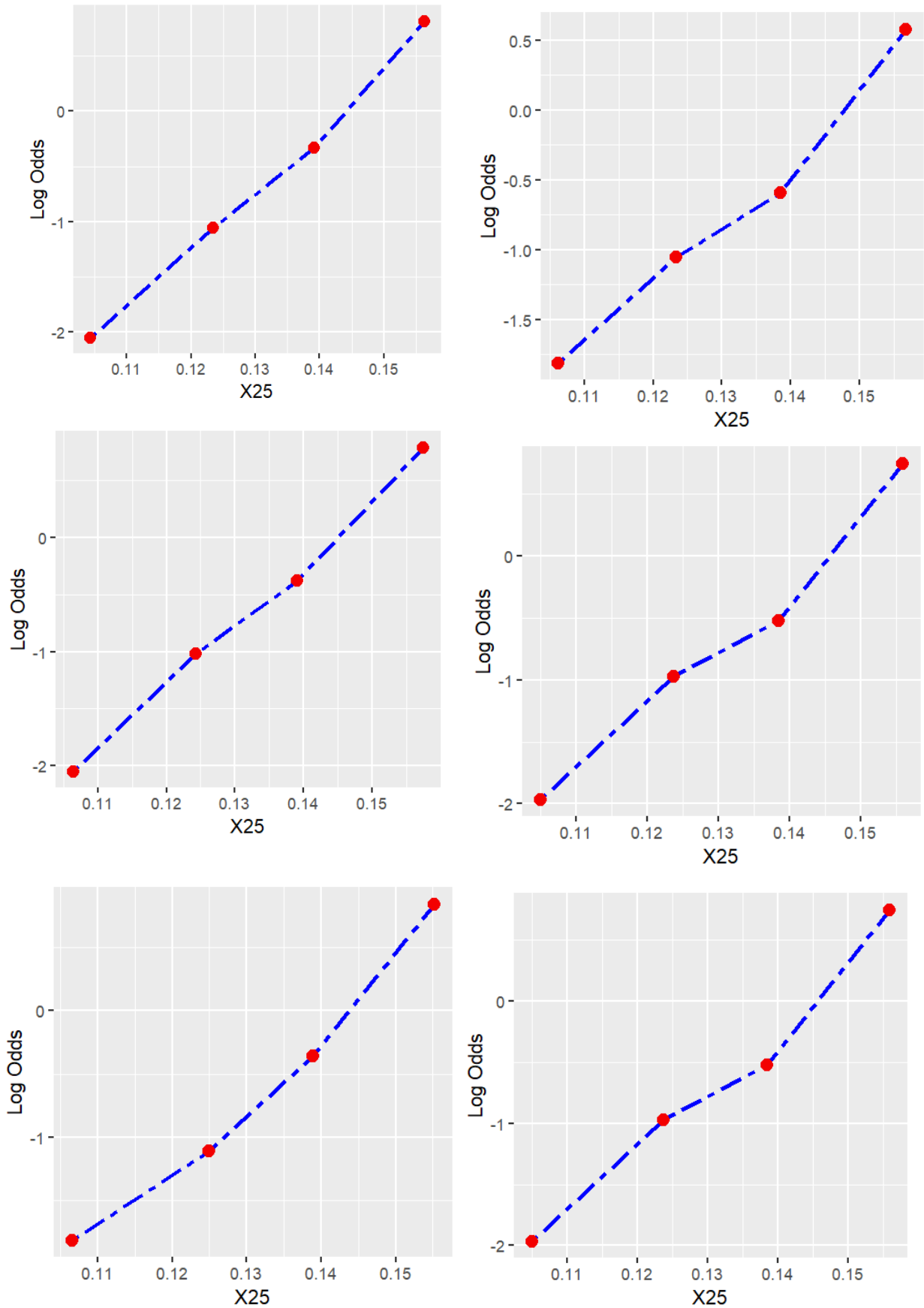
LINEARITY CHECK FOR RADIUS WORST(X21)



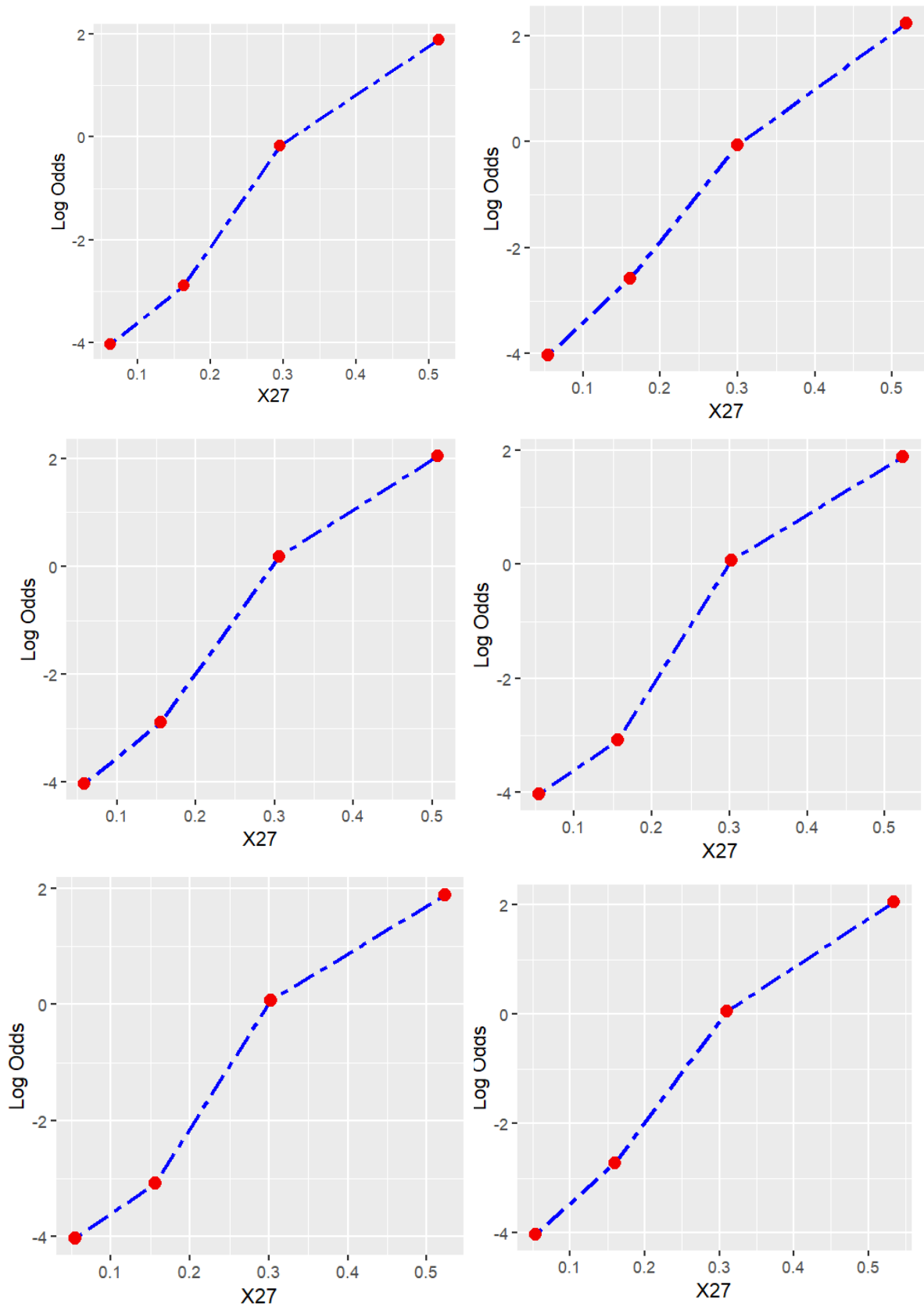
LINEARITY CHECK FOR TEXTURE WORST(X22)



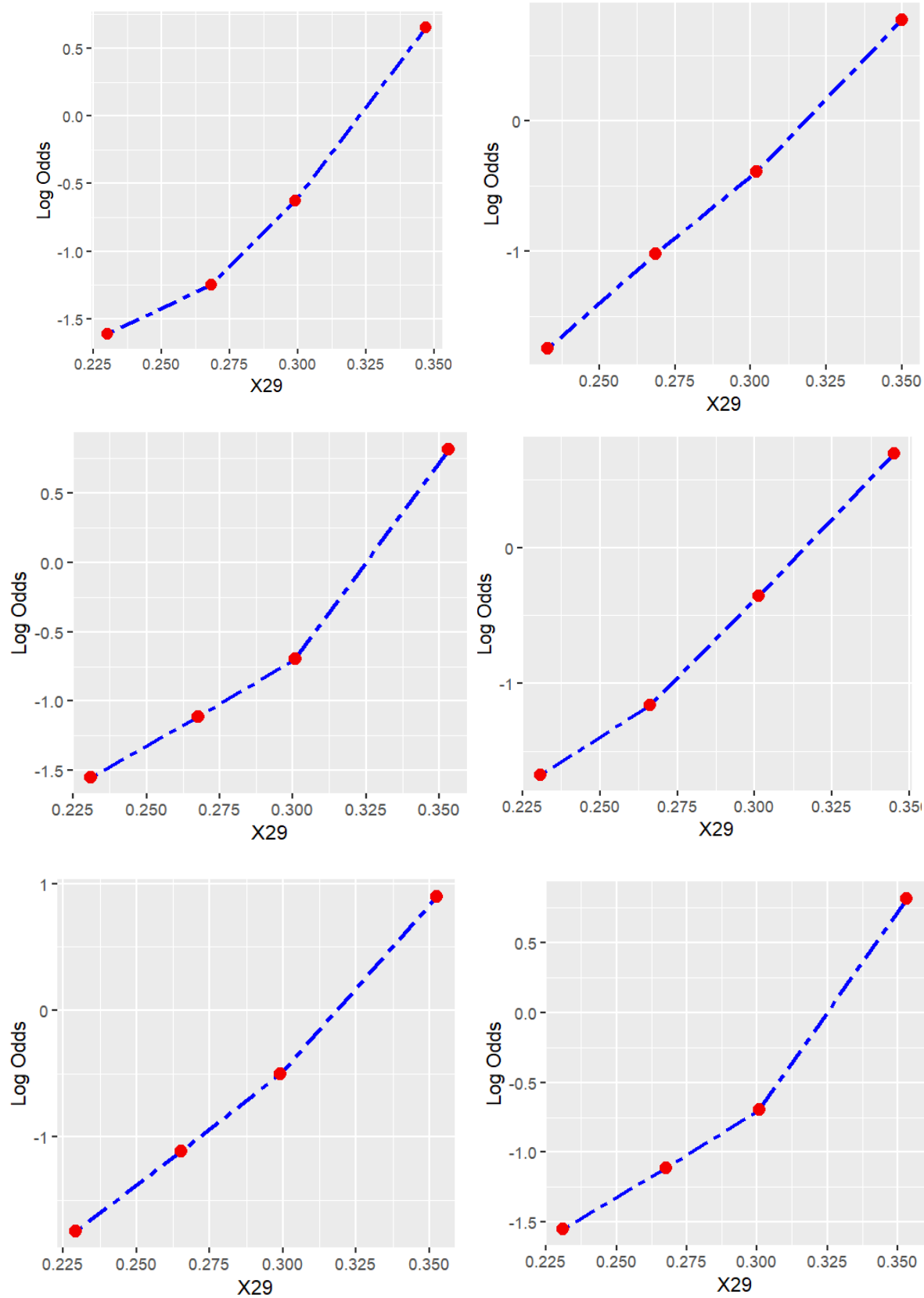
LINEARITY CHECK FOR SMOOTHNESS WORST(X25)



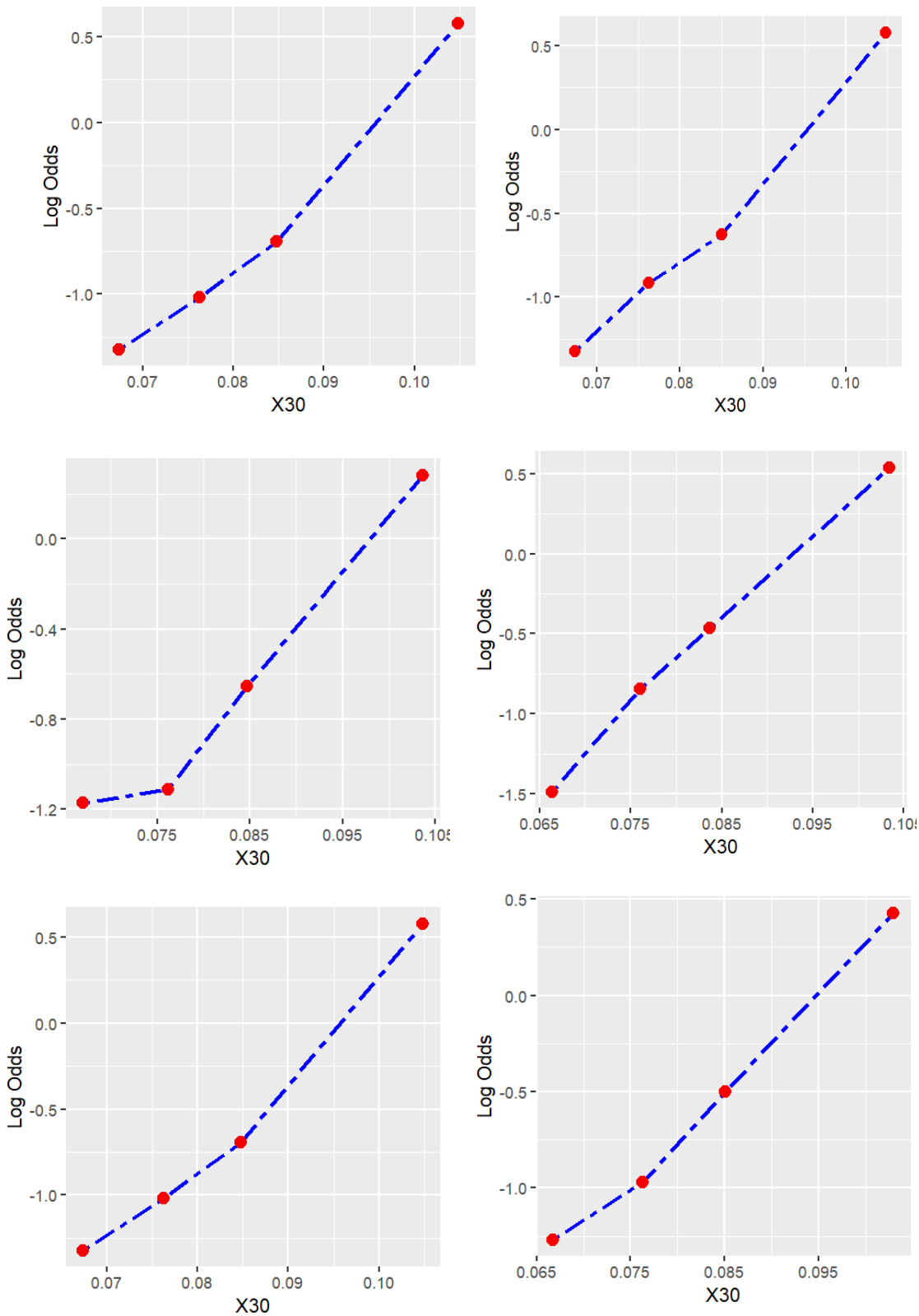
LINEARITY CHECK FOR CONCAVITY WORST(X27)



LINEARITY CHECK FOR SYMMETRY WORST(X29)



LINEARITY CHECK FOR FRACTAL DIMENSION WORST(X30)



8.6.4 Variable reduction- Phase II

After checking linearity for the remaining 18 variables (after phase-I reduction), we can see that *fractal dimension mean*(X10), *texture se*(X12), *smoothness se*(X15), *concavity se*(X16), *symmetry se*(X19), *fractal dimension se*(X20) are validating the linearity assumption of Logistic regression. So, we drop these 6 variables.

The testing dataset should be adjusted accordingly.

8. MODEL FITTING

After 2 stage variable reduction we have 12 independent predictor variables and the dependent response variables remaining. Now we fit logistic regression model to this dataset.

8.1) LOGISTIC REGRESSION:

Let us denote the response variable Malignancy by Y. Here Y is a binary variable denoting the malignant (breast cancer) condition of cells.

Here, $Y = 1$, if the tumour is malignant

0 , if the tumour is benign

We assume that, $Y_i \sim \text{Bernoulli}(\pi_i)$, $i = 1(1)n$

Where, π_i denotes the probability of the i-th person having malignancy (breast cancer) ($i = 1(1)n$) and n denote the total number of rows in the cleaned training dataset. Here, $n = 455$

8.1.1 Model based on mean, se and worst variables:

Here our regression equation is:

$$P(Y=1 | X_1, X_2, X_5, X_7, X_{11}, X_{21}, X_{22}, X_{25}, X_{27}, X_{29}, X_{30})$$

$$= \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_5 X_5 + \dots + \beta_{30} X_{30})}}$$

The logistic regression is the linear combination of predictors. And it models the log odds of probability (of having malignancy) with respect to different predictors.

MODEL:

$$Y = \log \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_5 X_5 + \dots + \beta_{30} X_{30} + \epsilon$$

In general,

$$Y_i = \log \left(\frac{p_i}{1-p_i} \right) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_5 X_{5i} + \dots + \beta_{30} X_{30i} + \epsilon_i, \quad i=1(1)n;$$

Here $\log \left(\frac{p_i}{1-p_i} \right)$ denotes the log odds of $Y_i = 1$

$p = P(Y = 1 | X_1, X_2, X_5, X_7, X_{11}, X_{21}, X_{22}, X_{25}, X_{27}, X_{29}, X_{30})$ is the probability of Malignancy;

β_0 is the intercept term;

X_1 is *Radius mean*;

X_2 is the *Texture mean*;

X_5 is the *Smoothness mean*;

X_7 is the *Concavity mean*;

X_9 is the *Symmetry mean*;

X_{11} is the *Radius se*;

X_{21} is the *Radius worst*;

X_{22} is the *Texture worst*;

X_{25} is the *Smoothness worst*;

X_{27} is the *Concavity worst*;

X_{29} is the *Symmetry worst*;

X_{30} is the *Fractal dimension worst*;

$\beta_1, \beta_2, \beta_5, \dots, \beta_{30}$ are the coefficients associated with the predictors respectively; indicating how much the log-odds of the outcome change with one-unit increase in each predictor;

And, ϵ is the random error associated to the model;

Parameter Estimation:

The parameters $\beta_1, \beta_2, \beta_5, \dots, \beta_{30}$ are estimated using Method of Maximum Likelihood.

The likelihood function for n observations is: $L(\beta) = \prod_{i=1}^n p_i^{Y_i} (1-p_i)^{(1-Y_i)}$

The log-likelihood function is then given by –

$$\log(L) = \sum_{i=1}^n Y_i * \ln p_i + (1 - Y_i) * \ln(1 - p_i)$$

The Score equations are given by,

$$\sum_{i=1}^n (Y_i - p_i) = 0$$

$$\sum_{i=1}^n (Y_i - p_i) X_{1i} = 0$$

$$\sum_{i=1}^n (Y_i - p_i) X_{2i} = 0$$

.

.

$$\sum_{i=1}^n (Y_i - p_i) X_{30i} = 0$$

We solve the score equations simultaneously and obtain the estimates $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2 \dots \hat{\beta}_{30}$.

Then we fit a logistic regression on the data and observe if there is still any presence of **Multicollinearity**.

Multicollinearity happens when two or more predictor variables within a regression model are correlated highly, which implies that they hold redundant or overlapping information. Multicollinearity makes it challenging to decide the effect of each predictor individually on the response variable. We use the `vif()` function from the “car” package in R to obtain the Variance Inflation Factor (VIF).

- Variance Inflation Factor (VIF) is a measure used in statistical regression analysis to assess the severity of multicollinearity in a set of variables. VIF quantifies how much the variance of a regression coefficient is inflated due to multicollinearity.
- VIF for j-th predictor is given by $VIF_j = \frac{1}{1-R_j^2}$ is the R^2 value obtained from a regression of the j-th predictor against all other predictors.
- In general, $VIF > 5$ means multicollinearity exists. The Variance Inflation Factor (VIF) indicates to what extent the variance of a regression coefficient is inflated because of multicollinearity. That is, it informs us if a predictor is highly correlated with other predictors in the model.

We find the VIF score as,

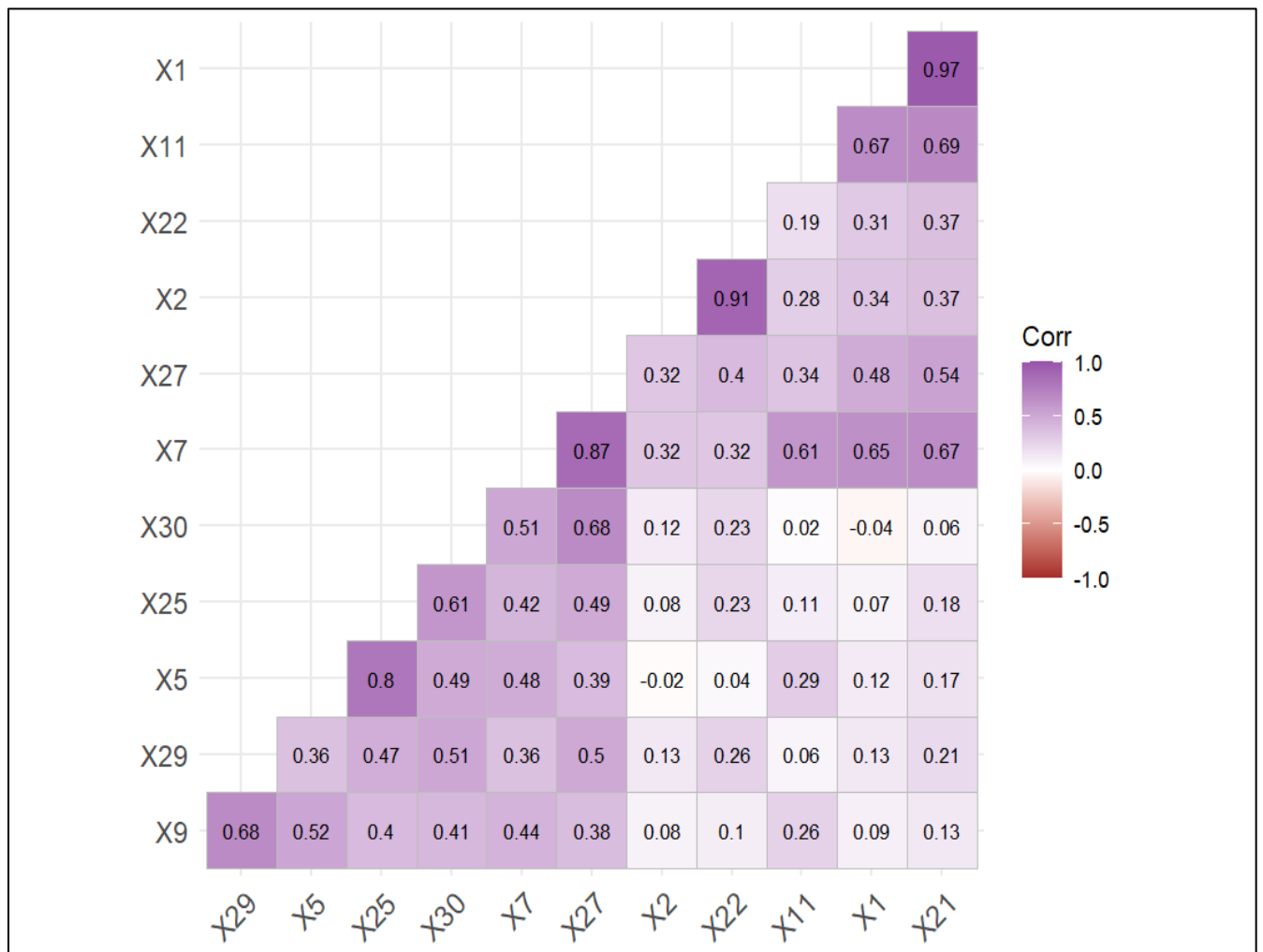
PREDICTORS	VIF SCORE
X1	22.478432 *
X2	6.652703 *
X5	5.346379
X7	7.756803 *
X9	3.929760
X11	3.276721
X21	22.647631*
X22	6.369618*
X25	6.246158*
X27	8.571505*
X29	4.006358
X30	3.400993

We can see that so many variables have high VIF score, Especially X1(Radius mean) and X22(Texture se). That means the model will not be good if we consider all the above mentioned variables.

So, we need to reconstruct the model.

Let us plot the correlation matrix for the variables.

CORRELATION MATRIX PLOT FOR THE PREDICTOR VARIABLES OF THE ABOVE MODEL:



From the plot we can observe that the *mean variables* and the *worst variables* are **highly correlated**. The worst variables capture less information as compared to mean variables; but many of the worst variables have strong linear relation with the log odds of the response variable. So, it will be unwise to drop any one group of variables.

Rather we split the Training Set into two parts; one having the *mean* and *se* variables, the other one having the *se* and *worst* variables. We fit models on both types of training sets and finally we shall stick with the model having better accuracy.

8.1.2 Model based on mean and se variables:

MODEL:

$$Y = \log \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_5 X_5 + \beta_7 X_7 + \beta_9 X_9 + \beta_{11} X_{11} + \epsilon_1$$

Where,

$p = P(Y = 1 | X_1, X_2, X_5, X_7, X_9, X_{11})$ is the probability of Malignancy;

β_0 is the intercept term;

X_1 is *Radius mean*;

X_2 is the *Texture mean*;

X_5 is the *Smoothness mean*;

X_7 is the *Concavity mean*;

X_{11} is the *Radius se*;

ϵ_1 is the random error in the model.

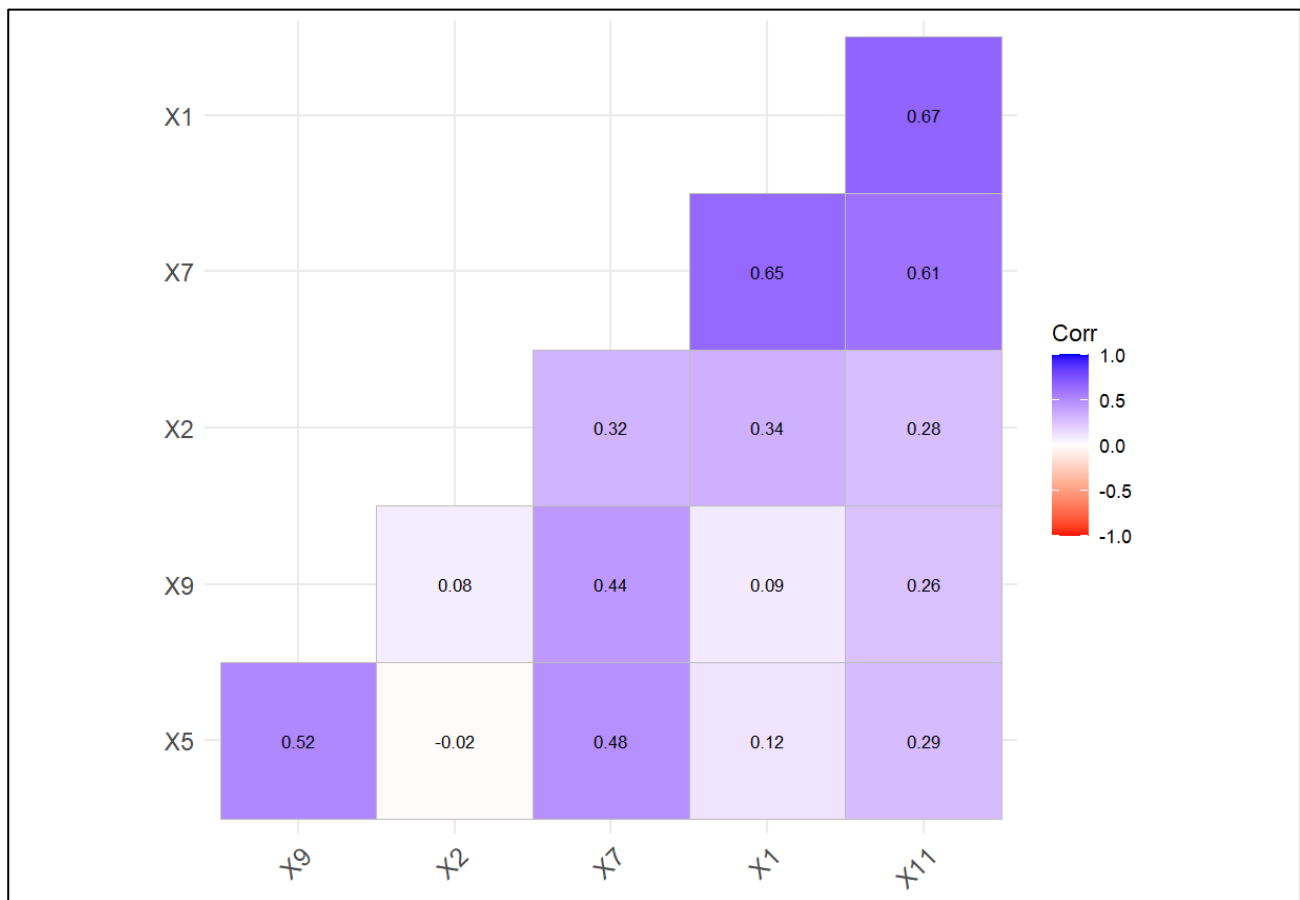
The parameter estimation process is same as before.

Now the new VIF scores are,

PREDICTORS	VIF SCORE
X1	1.859854
X2	1.688560
X5	2.324231
X7	1.360341
X9	1.849384
X11	1.118864

All the VIF s are pretty small, the maximum being 2.324231; so we are ensured that multi-collinearity is not a significant concern among the predictor variables in our regression model. Thus we have successfully omitted the problem of multicollinearity.

Correlation matrix for the visualization of the predictors:



```
> summary(fit_logistic_ms)
```

Call:

```
glm(formula = Y ~ X1 + X2 + X5 + X7 + X9 + X11, family = binomial(link = "logit"),
    data = traindata_set1)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-39.58078	5.46392	-7.244	4.36e-13	***
X1	1.22612	0.17419	7.039	1.94e-12	***
X2	0.37139	0.07051	5.267	1.39e-07	***
X5	102.08473	23.91359	4.269	1.96e-05	***
X7	16.37013	4.29153	3.815	0.000136	***
X9	14.79579	11.60132	1.275	0.202184	
X11	0.45026	1.37811	0.327	0.743876	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 588.87 on 454 degrees of freedom
 Residual deviance: 132.07 on 448 degrees of freedom
 AIC: 146.07

Number of Fisher Scoring iterations: 8

So, by Fisher Scoring, we get the estimates of the model parameters.

β_0	-39.58078
β_1	1.22612
β_2	0.37139
β_5	102.08473
β_7	16.37013
β_9	14.79579
β_{11}	0.45026

So, the fitted model is given by,

$$\text{Log}\left(\frac{p}{1-p}\right) = -39.58078 + 1.22612 X_1 + 0.37139 X_2 + 102.08473 X_5 + 16.37013 X_7 + 14.79579 X_9 + 0.45026 X_{11}$$

INTERPRETATION OF THE MODEL PARAMETERS:

$\hat{\beta}_0 = -39.58078$ means when all the predictors are zero, the log odds of malignancy is -39.58078 times the log odds of non- malignancy.

$\hat{\beta}_1 = 1.22612$ means keeping other factors fixed, a one unit increase in *radius mean* increases the log odds of malignancy by 1.22612.

$\hat{\beta}_2 = 0.37139$ means keeping other factors fixed, a one unit increase in *texture mean* increases the log odds of malignancy by 0.37139.

$\hat{\beta}_5 = 102.08473$ means keeping other factors fixed, a one unit increase in *smoothness mean* increases the log odds of malignancy by 102.08473.

$\hat{\beta}_7 = 16.37013$ means keeping other factors fixed, a one unit increase in *concavity mean* increases the log odds of malignancy by 16.37013.

$\hat{\beta}_9 = 14.79597$ means keeping other factors fixed, a one unit increase in *symmetry mean* increases the log odds of malignancy by 14.79597.

$\hat{\beta}_{11} = 0.45026$ means keeping other factors fixed, a one unit increase in *radius se* increases the log odds of malignancy by 0.45026.

8.1.3. Model based on se and worst variables:

MODEL:

$$Y = \log \left(\frac{p}{1-p} \right) = \beta_0 + \beta_{11} X_{11} + \beta_{21} X_{21} + \beta_{22} X_{22} + \beta_{25} X_{25} + \beta_{27} X_{27} + \beta_{29} X_{29} + \beta_{30} X_{30} + \epsilon_2$$

Where,

$p = P(Y = 1 | X_{11}, X_{21}, X_{22}, X_{25}, X_{27}, X_{29}, X_{30})$ is the probability of Malignancy;

β_0 is the intercept term;

X_{11} is the *Radius se*;

X_{21} is the *Radius worst*;

X_{22} is the *Texture worst*;

X_{25} is the *Smoothness worst*;

X_{27} is the *Concavity worst*;

X_{29} is the *Symmetry worst*;

X_{30} is the *Fractal dimension worst*;

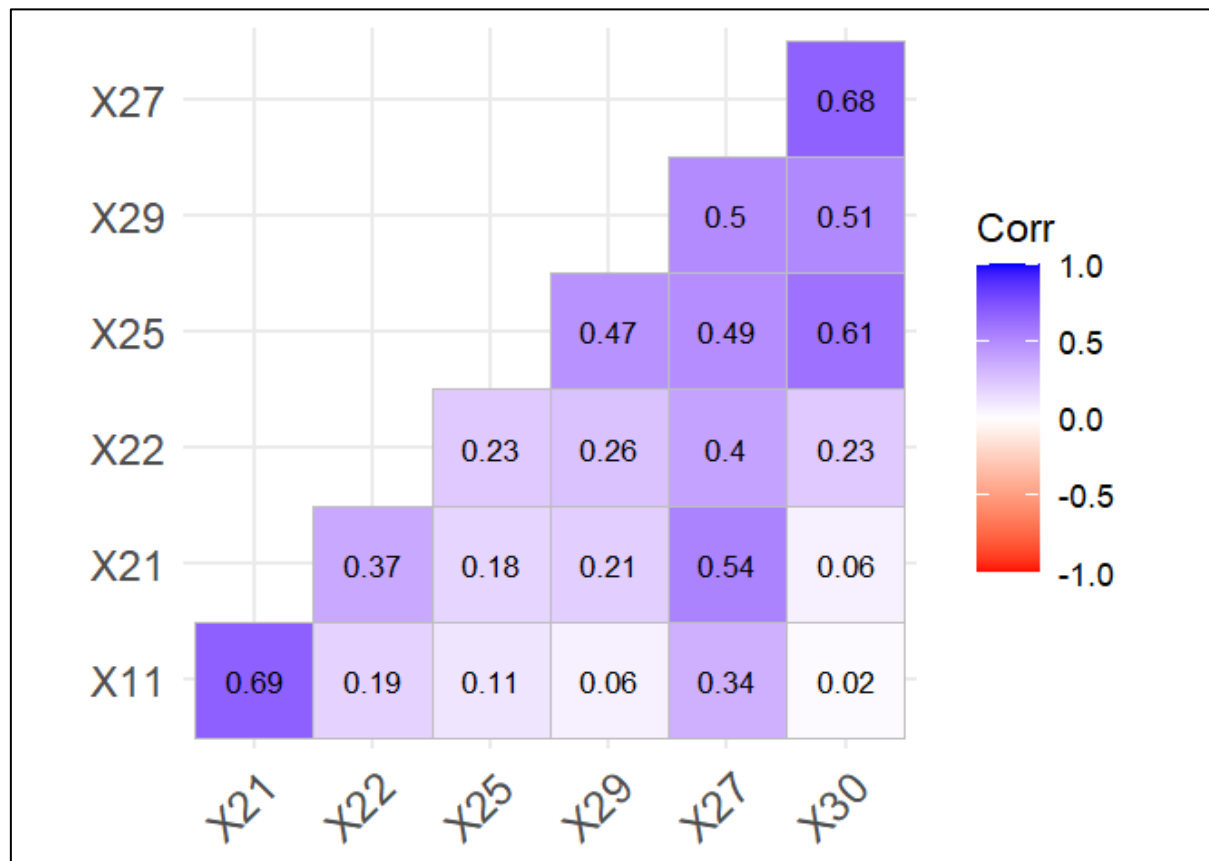
ϵ_2 is the random error in the model.

The parameter estimation process is same as before.

Now the new VIF scores are,

PREDICTORS	VIF SCORES
X_{11}	2.098298
X_{21}	2.000603
X_{22}	1.581062
X_{25}	2.530796
X_{27}	2.181146
X_{29}	1.660730
X_{30}	2.811888

Here also all the VIF s are pretty small, the maximum being 2.811888; So we are ensured that multi-collinearity is not a significant concern among the predictor variables in our regression model

Correlation matrix for the visualization of the predictors:

```
> summary(fit_logistic_sw)
```

Call:

```
glm(formula = Y ~ X11 + X21 + X22 + X25 + X27 + X29 + X30, family = binomial(link = "logit"),
    data = traindata_set1)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-50.00061	8.70475	-5.744	9.24e-09	***
X11	10.21825	2.48428	4.113	3.90e-05	***
X21	1.48365	0.27201	5.454	4.91e-08	***
X22	0.30143	0.07074	4.261	2.03e-05	***
X25	83.31032	23.39220	3.561	0.000369	***
X27	7.45417	2.67001	2.792	0.005241	**
X29	13.32311	6.19828	2.149	0.031596	*
X30	-35.81869	35.73333	-1.002	0.316156	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 588.866 on 454 degrees of freedom
 Residual deviance: 64.506 on 447 degrees of freedom
 AIC: 80.506

Number of Fisher Scoring iterations: 9

So, by Fisher Scoring, we get the estimates of the model parameters.

β_0	-50.00061
β_{11}	10.21825
β_{21}	1.48365
β_{22}	0.30143
β_{25}	83.31032
β_{27}	7.45417
β_{29}	13.32311
β_{30}	-35.81869

So, the fitted model is given by,

$$\text{Log}\left(\frac{p}{1-p}\right) = -50.00061 + 10.21825X_{11} + 1.48365X_{21} + 0.30143X_{22} + 83.31032X_{25} + 7.45417X_{27} + 13.32311X_{29} - 35.81869X_{30}$$

INTERPRETATION OF THE MODEL PARAMETERS:

$\hat{\beta}_0 = -50.00061$ means when all the predictors are zero, the log odds of malignancy is -50.00061 times the log odds of non-malignancy.

$\hat{\beta}_{11} = 10.2185$ means keeping other factors fixed, a one unit increase in *radius se* increases the log odds of malignancy by 1.22612.

$\hat{\beta}_{21} = 1.48365$ means keeping other factors fixed, a one unit increase in *radius worst* increases the log odds of malignancy by 1.48365.

$\hat{\beta}_{22} = 0.30143$ means keeping other factors fixed, a one unit increase in *texture worst* increases the log odds of malignancy by 0.30143.

$\hat{\beta}_{25} = 83.31032$ means keeping other factors fixed, a one unit increase in *smoothness worst* increases the log odds of malignancy by 83.31032.

$\hat{\beta}_{27} = 7.45417$ means keeping other factors fixed, a one unit increase in *concavity worst* increases the log odds of malignancy by 7.45417.

$\hat{\beta}_{29} = 13.32311$ means keeping other factors fixed, a one unit increase in *symmetry se* increases the log odds of malignancy by 13.32311.

$\hat{\beta}_{30} = -35.81869$ means keeping other factors fixed, a one unit increase in *symmetry se* increases the log odds of malignancy by -35.81869.

9. GOODNESS OF FIT

9.1) COMPARING NULL DEVIANCE AND RESIDUAL DEVIANCE:

❖ Null Deviance:

The null deviance reflects the goodness of fit of a model that only includes the intercept (the null model), meaning it has no predictors. It shows the difference in deviance between the null model and the model being assessed. A lower null deviance suggests that the model accounts for a greater proportion of the variance in the response variable compared to the null model.

❖ Residual Deviance:

The residual deviance indicates the goodness of fit of the model after adding predictors. It represents the difference in deviance between the evaluated model and the saturated model (which perfectly fits the data by having a parameter for each observation). A smaller residual deviance signifies a better fit of the model to the data.

The deviance residual for i -th observation is given by-

$$d_i = \pm [-2 \{Y_i \log \hat{p}_i + (1 - Y_i) \log(1 - \hat{p}_i)\}]^{1/2}$$

where the sign is positive when $Y_i \geq \hat{p}_i$ and negative otherwise;

$$Y_i = \begin{cases} 1, & \text{If } i\text{-th observation is malignant} \\ 0, & \text{If } i\text{-th observation is benign} \end{cases}$$

and $\hat{p}_i$ is the predicted probability for i -th observation.

The residual deviance statistic for a model is given by $D = \sum_i d_i^2$

For Model-1 (based on mean and se variables):

After fitting the logistic model in R, we get that-

Null deviance: 588.87 on 454 degrees of freedom

Residual deviance: 132.07 on 448 degrees of freedom

This suggest that the predictors included in the model explain a significant proportion of the variance in the response variable. So, the fitted model is a good fit.

For Model-2(based on mean and se variables):

After fitting the logistic model in R, we get that-

Null deviance: 588.866 on 454 degrees of freedom

Residual deviance: 64.506 on 447 degrees of freedom

This suggest that the predictors included in the model explain a significant proportion of the variance in the response variable. So, the fitted model is a good fit.

9.2) COMPARING QUALITY EVALUATION OF THE MODEL:

We evaluate the quality of the logistic regression model by forming a Confusion Matrix.

Definition of a Confusion Matrix:

A confusion matrix is a performance evaluation tool used in classification models to assess how well the model distinguishes between different classes. It provides a 2x2 tabular summary of predictions by comparing actual and predicted values, offering insights into the types of errors the model makes. By breaking down predictions into categories such as **true positive rate (TPR)**, **false positive rate (FPR)**, the confusion matrix helps in calculating key evaluation metrics like accuracy, precision etc. This detailed breakdown enables a deeper understanding of model performance beyond a single accuracy measure.

Different evaluation metrics are given as follows:

- (1) **Accuracy** indicates how many labels out of the total labels are predicted correctly by a classification model.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

Where,

- **True Positive (TP)** is the actual class of the observations that are Benign and the model also predicts it as Benign.
- **True Negative (TN)** is the actual class of the observations that are Malignant and the model also predicts it as Malignant.
- **False Positive (FP)** is the actual class of the observations that are Malignant but the model predicts it incorrectly as Benign.
- **False Negative (FN)** is the actual class of the observations that are Benign but the model predicts it incorrectly as Malignant.

- (2) **Recall** is also known as sensitivity and can be defined as the ratio of the total number of correctly predicted positive examples to the total number of positive examples. A high value of recall indicates that the class is correctly recognized (a small number of FNs).

$$\text{Recall} = \frac{TP}{TP+FN}$$

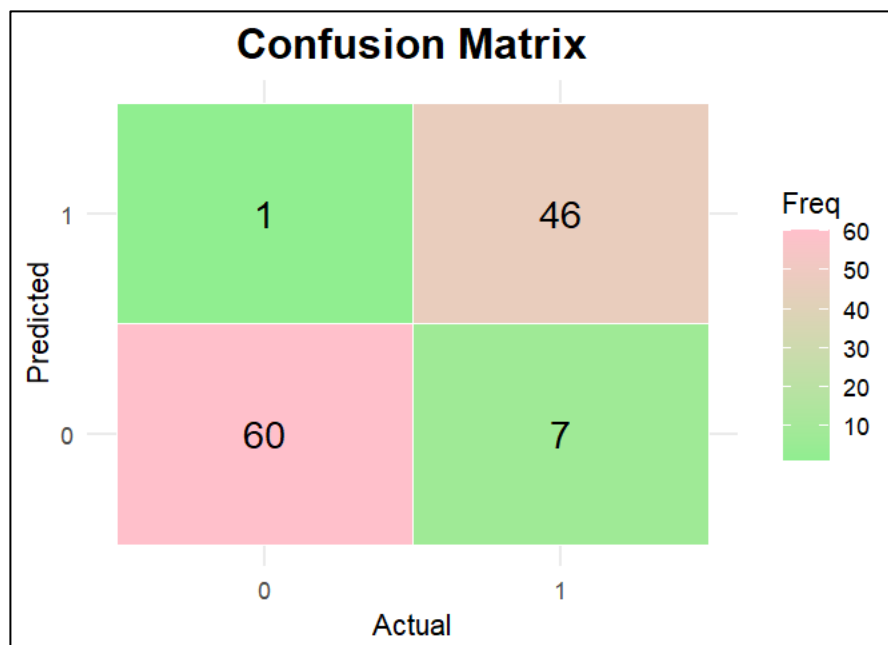
- (3) **Precision** is also known as the exactness of classifiers. It is a measure of how accurate aa model's positive predictions are.

$$\text{Precision} = \frac{TP}{TP+FP}$$

- (4) **F1 Score** is also known as the F measure and it is the harmonic mean of the precision and recall scores. It is a metric that provides a balanced evaluation of a model's performance, particularly useful when dealing with imbalanced datasets.

$$\text{F1 Score} = \frac{2 * \text{Precision} * \text{Recall}}{(\text{Precision} + \text{Recall})}$$

9.2.1 Confusion matrix for the Model-1(model with mean and se variables):

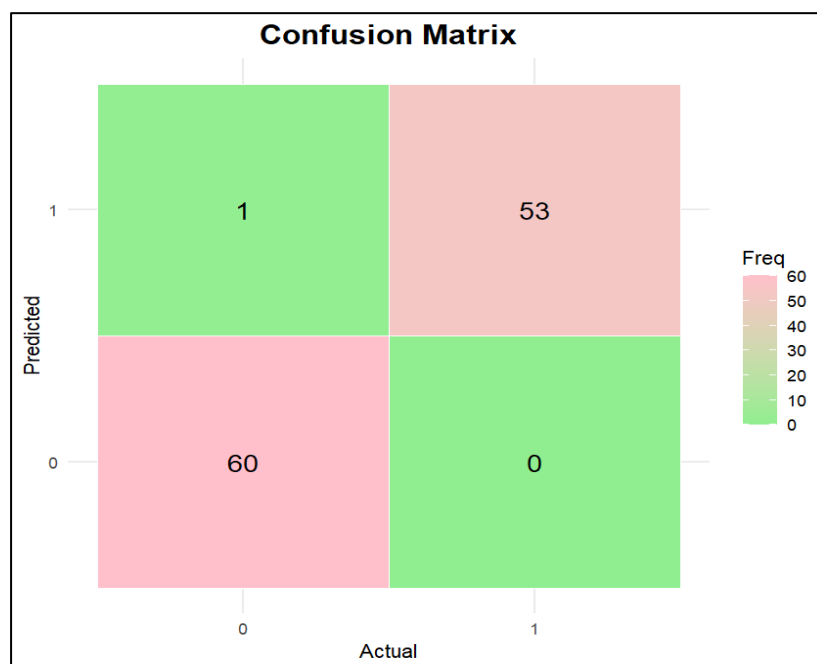


Here, TP = 60; TN = 46; FP= 7; FN= 1 [Here “0” means Benign, “1” means Malignant]

So, for the model with mean and se variables –

- a. **Accuracy** is 0.92982
- b. **Recall** is 0.98361
- c. **Precision** is 0.89552
- d. **F1 score** is 0.93750

9.2.2 Confusion matrix for the Model-2(model with se and worst variables):



Here, TP = 60; TN = 53; FP= 0; FN=1 [Here “0” means Benign, “1” means Malignant]

So, for the model with mean and se variables –

- a. **Accuracy** is 0.99123
- b. **Recall** is 0.98361
- c. **Precision** is 1
- d. **F1 score** is 0.99174

9.3) BETTER FITTED MODEL:

A. Less residual deviance:

Residual deviance for the model with *mean* and *se* variables is **132.07** and for the model with *se* and *worst* variables is **64.506**. Clearly, the latter provides a better fit.

B. More Accuracy:

We compare the two models by their respective accuracies. In order to do so, we train our models on multiple training sets, then run them on the corresponding test sets and note the accuracies.

For the model with mean and se variables (Model 1), the accuracies are –

```
> print(results_accuracy)
Run train_size test_size Accuracy
1 1 455 114 0.9386
2 2 455 114 0.9298
3 3 455 114 0.9474
4 4 455 114 0.9211
5 5 455 114 0.9561
> cat("Mean Accuracy across", n_runs, "runs:", round(mean(results_accuracy$Accuracy),4),"\n")
Mean Accuracy across 5 runs: 0.9386
```

For the model with se and worst variables (Model 2), the accuracies are –

```
> print(results_accuracy)
Run train_size test_size Accuracy
1 1 455 114 0.9912
2 2 455 114 0.9561
3 3 455 114 0.9825
4 4 455 114 0.9649
5 5 455 114 0.9912
> cat("Mean Accuracy across", n_runs, "runs:", round(mean(results_accuracy$Accuracy),4),"\n")
Mean Accuracy across 5 runs: 0.9772
```

So, we can clearly see that for model 2 mean accuracy is greater than model 1.

That means, Model 2 (The model with se and worst variables) has better performance. Therefore, it will be better to use Model 2 for predicting Malignancy.

10. CONCLUSION

The aim of this project work was to study the potential of logistic regression in predicting breast cancer malignancy using cytological features derived from fine needle aspiration (FNA) tests. Through **data preprocessing, variable reduction, model fitting, and evaluation**, the project work achieved high levels of prediction accuracy and thus has fulfilled its aim.

To improve the model's performance and interpretability, we carefully reduced highly correlated variables to **address multicollinearity issues** and checked **whether the predictors maintained a linear relationship** with the log-odds of the outcome. This step-by-step selection approach helped us focusing on the most meaningful variables for diagnosis.

Here two logistic regression models were developed with different predictor variables and comparing this 2 we can see:

Model 1 (Mean + SE variables): It gives high recall and F1 score, good performance with mean accuracy 93.86% and residual deviance 132.07.

Model 2 (SE + Worst variables): It gives better performance with higher mean accuracy 97.72 % and a lower residual deviance 64.506.

Hence, **Model 2** is identified as the better diagnostic tool, offering a balance of simplicity, interpretability, and performance. For Model 2 the predictor variables are 1) Radius se, 2) Radius worst, 3) Texture worst, 4) Smoothness worst, 5) Concavity worst 6) Symmetry worst 7) Fractal dimension worst. It effectively distinguishes between benign and malignant cases with minimal error.

Despite the success of the model, it is to be noted that this logistic regression model should not be used as a standalone diagnostic method. Instead, it can serve as a supplementary decision support tool to assist clinicians in early detection and screening, particularly in resource-constrained settings.

In the bigger picture, this project shows how even simple statistical models, when built thoughtfully, can contribute meaningfully to medical diagnostics. It reinforces the value of integrating **data-driven approaches** in healthcare while prioritizing **transparency and reliability** in model interpretation.

11. REFERENCES

1. **Statistical Methods: Including Probability** – Goon, Gupta & Dasgupta
2. **Fundamentals of Mathematical Statistics** – S.C. Gupta & V.K. Kapoor
3. **Statistical Methods for Practice and Research** – Ajai S. Gaur, Sanjaya S. Gaur
4. **Logistic Regression: A Self-Learning Text** – David G. Kleinbaum, Mitchel Klein
5. **Applied Logistic Regression** – David W. Hosmer, Stanley Lemeshow, Rodney X. Sturdivant
6. **Applied Regression Analysis and Generalized Linear Models** – John Fox

12. APPENDIX

#Loading the dataset

```
Data= read.csv("C://Users//anwes//Downloads//breast-cancer (1).csv")
```

#Data dimension

```
dim(Data)
```

#Showing variables names

```
names(Data)
```

#Finding missing values

```
which(is.na(Data))
```

#Encoding

```
Data$diagnosis [data$diagnosis=="M"] =1
```

```
Data$diagnosis [data$diagnosis=="B"] = 0
```

#Data summary

```
summary(Data)
```

#Changing the name of the variables

```
DATA=data.frame(data)
```

```
names(Data)=c("Y","X1","X2","X3","X4","X5","X6",  
"X7","X8","X9","X10","X11","X12","X13","X14","X15","X16","X17","X18","X19","X20","X21","X22","X23",  
"X24","X25","X26","X27","X28","X29","X30")
```

#Splitting into training and testing dataset

```
library(caTools)
```

```
library(RTools)
```

```
split = sample.split(Data, SplitRatio = 0.8)
```

```
train_data = subset(Data, split=="TRUE")
```

```
test_data = subset(Data, split=="FALSE")
```

#Barplot of diagnosis for training dataset

```
table(train_data)
```

```
barplot(table(train_data$Y), main= "Barplot of diagnosis for Training dataset", xlab="Diagnosis", ylab=  
"Count", col=c("skyblue", "pink"), names.arg = c("Benign", "Malignant"))
```

#Correlation matrix for all the 30 predictor variables

```
library(corrplot)
```

```
library(ggcorrplot)
```

```
train_data = train_data[, 2:31]
```

```
cor_matrix = cor(train_data, use= "complete.obs")
```

```
ggcorrplot(cor_matrix, hc.order = TRUE, type= "lower", lab=2, lab_size=3, method = "square" , color= c("red",
"white", "blue"))
```

#KDE and pairwise scatter-plot for mean variables

```
library(GGally)
library(ggplot2)
library(dplyr)
train_data$Y = factor(train_data$Y)
train_data_subset = train_data[ , c(1, 2:11)]
ggpairs(train_data_subset,
        columns = 2:11,
        mapping = aes( color = Y, alpha= 0.6),
        diag = list(continuous = "densityDiag"), # KDE on diagonal
        upper = list(continuous = "points"),    # Scatter in upper
        lower = list(continuous = "points")) +   # Scatter in lower
        theme_minimal()
)
```

Reducing columns (phase-I)

```
train_data_set<- train_data %>% select (-X3, -X4, -X6, -X8, -X13, -X14, -X16, -X18, -X23, -X24, -X26, -X28)
```

Function for Linearity assumption checking

```
linearity_check <- function(data_set , variable_name){
  values <- data_set[[variable_name]]
  q <- quantile(values)
  g1 <- data_set$Y[which(values < q[2])]
  g2 <- data_set$Y[which(values >= q[2] & values < q[3])]
  g3 <- data_set$Y[which(values >= q[3] & values < q[4])]
  g4 <- data_set$Y[which(values >= q[4])]
  probs <- c()
  probs[1] <- sum(g1) / length(g1)
  probs[2] <- sum(g2) / length(g2)
  probs[3] <- sum(g3) / length(g3)
  probs[4] <- sum(g4) / length(g4)
  log_odds <- log(probs / (1- probs))
  med_1 <- median(values[which(values < q[2])])
  med_2 <- median(values[which(values >= q[2] & values < q[3])])
  med_3 <- median(values[which(values >= q[3] & values < q[4])])
}
```

```

med_4 <- median(values[which(values >= q[4])])
medians <- c(med_1, med_2, med_3, med_4)
temp_df <- data.frame(group_medians = medians, logodds = log_odds)
library(tidyverse)
line_graph <- temp_df %>%
  ggplot(aes(x = group_medians, y = logodds)) +
  geom_line(col = "blue", linewidth = 1, linetype = "twodash") +
  geom_point(size = 3, col = "#F30000") +
  labs(x = variable_name, y = "Log Odds")
return(line_graph)
}

```

Running the function for different training sets

```

training_sets <- list(train_data_set)
testing_sets <- list(test_data_set)
for (i in 1:5) {
  set.seed(runif(1, 1, 10))
  a <- sample(nrow(Data), nrow(Data) * 0.8)
  train_temp <- Data[a, ]
  test_temp <- Data[-a, ]
  training_sets[[i + 1]] <- train_temp
  testing_sets[[i + 1]] <- test_temp
}
linearity_check(training_sets[[1]], "X1")
linearity_check(training_sets[[2]], "X1")
linearity_check(training_sets[[3]], "X1")
linearity_check(training_sets[[4]], "X1")
linearity_check(training_sets[[5]], "X1")
linearity_check(training_sets[[6]], "X1")
linearity_check(training_sets[[1]], "X2")
linearity_check(training_sets[[2]], "X2")
linearity_check(training_sets[[3]], "X2")
linearity_check(training_sets[[4]], "X2")
linearity_check(training_sets[[5]], "X2")
linearity_check(training_sets[[6]], "X2")

```

```
linearity_check(training_sets[[1]], "X5")
linearity_check(training_sets[[2]], "X5")
linearity_check(training_sets[[3]], "X5")
linearity_check(training_sets[[4]], "X5")
linearity_check(training_sets[[5]], "X5")
linearity_check(training_sets[[6]], "X5")
linearity_check(training_sets[[1]], "X7")
linearity_check(training_sets[[2]], "X7")
linearity_check(training_sets[[3]], "X7")
linearity_check(training_sets[[4]], "X7")
linearity_check(training_sets[[5]], "X7")
linearity_check(training_sets[[6]], "X7")
linearity_check(training_sets[[1]], "X7")
linearity_check(training_sets[[1]], "X9")
linearity_check(training_sets[[2]], "X9")
linearity_check(training_sets[[3]], "X9")
linearity_check(training_sets[[4]], "X9")
linearity_check(training_sets[[5]], "X9")
linearity_check(training_sets[[6]], "X9")
linearity_check(training_sets[[1]], "X10")
linearity_check(training_sets[[2]], "X10")
linearity_check(training_sets[[3]], "X10")
linearity_check(training_sets[[4]], "X10")
linearity_check(training_sets[[5]], "X10")
linearity_check(training_sets[[6]], "X10")
linearity_check(training_sets[[5]], "X10")
linearity_check(training_sets[[4]], "X10")
linearity_check(training_sets[[3]], "X10")
linearity_check(training_sets[[5]], "X10")
linearity_check(training_sets[[6]], "X10")
linearity_check(training_sets[[1]], "X11")
linearity_check(training_sets[[2]], "X11")
linearity_check(training_sets[[3]], "X11")
linearity_check(training_sets[[4]], "X11")
```

```
linearity_check(training_sets[[5]], "X11")
linearity_check(training_sets[[6]], "X11")
linearity_check(training_sets[[1]], "X12")
linearity_check(training_sets[[2]], "X12")
linearity_check(training_sets[[3]], "X12")
linearity_check(training_sets[[4]], "X12")
linearity_check(training_sets[[5]], "X12")
linearity_check(training_sets[[6]], "X12")
linearity_check(training_sets[[1]], "X15")
linearity_check(training_sets[[2]], "X15")
linearity_check(training_sets[[3]], "X15")
linearity_check(training_sets[[4]], "X15")
linearity_check(training_sets[[5]], "X15")
linearity_check(training_sets[[6]], "X15")
linearity_check(training_sets[[1]], "X17")
linearity_check(training_sets[[2]], "X17")
linearity_check(training_sets[[3]], "X17")
linearity_check(training_sets[[4]], "X17")
linearity_check(training_sets[[5]], "X17")
linearity_check(training_sets[[6]], "X17")
linearity_check(training_sets[[1]], "X19")
linearity_check(training_sets[[1]], "X19")
linearity_check(training_sets[[2]], "X19")
linearity_check(training_sets[[3]], "X19")
linearity_check(training_sets[[4]], "X19")
linearity_check(training_sets[[5]], "X19")
linearity_check(training_sets[[6]], "X19")
linearity_check(training_sets[[1]], "X19")
linearity_check(training_sets[[1]], "X20")
linearity_check(training_sets[[2]], "X20")
linearity_check(training_sets[[3]], "X20")
linearity_check(training_sets[[4]], "X20")
linearity_check(training_sets[[5]], "X20")
linearity_check(training_sets[[6]], "X20")
```

```
linearity_check(training_sets[[1]], "X21")
linearity_check(training_sets[[2]], "X21")
linearity_check(training_sets[[3]], "X21")
linearity_check(training_sets[[2]], "X21")
linearity_check(training_sets[[4]], "X21")
linearity_check(training_sets[[5]], "X21")
linearity_check(training_sets[[6]], "X21")
linearity_check(training_sets[[1]], "X22")
linearity_check(training_sets[[2]], "X22")
linearity_check(training_sets[[3]], "X22")
linearity_check(training_sets[[4]], "X22")
linearity_check(training_sets[[5]], "X22")
linearity_check(training_sets[[6]], "X22")
linearity_check(training_sets[[1]], "X25")
linearity_check(training_sets[[2]], "X25")
linearity_check(training_sets[[3]], "X25")
linearity_check(training_sets[[4]], "X25")
linearity_check(training_sets[[5]], "X25")
linearity_check(training_sets[[6]], "X25")
linearity_check(training_sets[[1]], "X27")
linearity_check(training_sets[[2]], "X27")
linearity_check(training_sets[[3]], "X27")
linearity_check(training_sets[[4]], "X27")
linearity_check(training_sets[[5]], "X27")
linearity_check(training_sets[[6]], "X27")
linearity_check(training_sets[[1]], "X29")
linearity_check(training_sets[[2]], "X29")
linearity_check(training_sets[[3]], "X29")
linearity_check(training_sets[[4]], "X29")
linearity_check(training_sets[[5]], "X29")
linearity_check(training_sets[[6]], "X29")
linearity_check(training_sets[[1]], "X30")
linearity_check(training_sets[[2]], "X30")
linearity_check(training_sets[[3]], "X30")
```

```
linearity_check(training_sets[[4]], "X30")
```

```
linearity_check(training_sets[[5]], "X30")
```

```
linearity_check(training_sets[[6]], "X30")
```

#Reducing columns(phase-II)

```
traindata_set1<- train_data_set %>% select (-X10, -X12, -X15, -X17, -X19, -X20)
```

Fitting logistic model for mean, se and worst variables

```
fit_logistic<-glm(Y ~ X1+X2+X5+X7+X9+X11+X21+X22+X25+X27+X29+X30,
```

```
data=traindata_set1,family=binomial(link = "logit"))
```

```
summary(fit_logistic) #summary of the model
```

```
library(car)
```

```
vif(fit_logistic) #showing VIF scores
```

#Fitting logistic model for mean and se variables

```
fit_logistic_ms<-glm(Y~X1+X2+X5+X7+X9+X11, data=traindata_set1,family=binomial(link = "logit"))
```

```
summary(fit_logistic_ms) #summary of the model
```

```
vif(fit_logistic_ms) #showing VIF scores
```

```
traindata_set1 = traindata_set1[ , 2: 7]
```

```
cor_matrix = cor(traindata_set , use= "complete.obs") #Correlation plot for the predictor variables
```

```
ggcorrplot(cor_matrix, hc.order = TRUE, type= "lower", lab=2, lab_size=3, method = "square" , color= c("red",  
"white", "blue"))
```

#Fitting logistic model for se and worst variables

```
fit_logistic_sw<-glm(Y~X11+X21+X22+X25+X27+X29+X30, data=traindata_set1,family=binomial(link =  
"logit"))
```

```
summary(fit_logistic_sw) # Summary of the model
```

```
vif(fit_logistic_sw) # Showing VIF scores
```

```
traindata_set1 = traindata_set1[ , 7: 13]
```

```
cor_matrix = cor(traindata_set , use= "complete.obs") #Correlation plot for the predictor variables
```

```
ggcorrplot(cor_matrix, hc.order = TRUE, type= "lower", lab=2, lab_size=3, method = "square" , color= c("red",  
"white", "blue"))
```

#Confusion matrix for model with mean and se variables

```
pred_prob <- predict (fit_logistic_ms, newdata = testdata_set1, type = "response")
```

```
pred_class <- ifelse(pred_prob > 0.5, 1, 0)
```

```
cm_table <- table(Predicted = predicted, Actual = actual)
```

```
ggplot(cm_df, aes(x = Actual, y = Predicted, fill = Freq)) +
```

```
  geom_tile(color = "white") +
```

```
  geom_text(aes(label = Freq), size = 5) +
```

```

scale_fill_gradient(low = "lightgreen", high = "pink") +
labs(title = "Confusion Matrix", x = "Actual", y = "Predicted") +
theme_minimal() + theme(plot.title = element_text(size = 16, face = "bold", hjust = 0.5))

#Confusion matrix for model with se and worst variables

pred_prob_sw<- predict(fit_logistic_sw, newdata = testdata_set2,type = "response")
pred_class_sw <- ifelse(pred_prob_sw > 0.5, 1, 0)
cm_table_sw <- table(Predicted = pred_class_sw, Actual = testdata_set2$Y)
cm_df_sw <- as.data.frame(cm_table_sw)

library(ggplot2)

ggplot(cm_df_sw, aes(x = Actual, y = Predicted, fill = Freq)) +
  geom_tile(color = "white") +
  geom_text(aes(label = Freq), size = 5) +
  scale_fill_gradient(low = "lightgreen", high = "pink") +
  labs(title = "Confusion Matrix", x = "Actual", y = "Predicted") +
  theme_minimal() +
  theme (plot.title = element_text(size = 16, face = "bold", hjust = 0.5))

#Mean Accuracy for Model-1

library(tidyverse)
library(pROC)

for(i in 1:n_runs){
  train_indices<- sample(1:nrow(df), size = floor(train_ratio * nrow(df)))
  train_data<-df[train_indices, ]
  test_data<-df[-train_indices, ]
  logistic_ms<- glm(Y ~ X1+ X2+ X5+ X7+ X9+ X11, data=train_data, family= "binomial")
  probs<- predict(logistic_ms, newdata = test_data, type= "response")
  preds<- ifelse(probs > 0.5, 1,0)
  actual<- as.numeric(as.character(test_data$Y))
  acc<- mean(preds== actual)
  results_accuracy$train_size[i]<- nrow(train_data)
  results_accuracy$test_size[i]<-nrow(test_data)
  results_accuracy$Accuracy[i]<- round(acc, 4)
}

print(results_accuracy)

cat("Mean Accuracy across", n_runs , "runs", round(mean(result_accuracy$Accuracy),4), "\n")

```

#Mean Accuracy for Model-2

```
for(i in 1:n_runs){  
  train_indices<- sample(1:nrow(df), size = floor(train_ratio * nrow(df)))  
  train_data<-df[train_indices, ]  
  test_data<-df[-train_indices, ]  
  logistic_sw<- glm(Y ~ X11+X21+X22+X25+X27+X29+X30, data=train_data, family= "binomial")  
  probs<- predict(logistic_ms, newdata = test_data, type= "response")  
  preds<- ifelse(probs > 0.5, 1,0)  
  actual<- as.numeric(as.character(test_data$Y))  
  acc<- mean(preds== actual)  
  results_accuracy$train_size[i]<- nrow(train_data)  
  results_accuracy$test_size[i]<-nrow(test_data)  
  results_accuracy$Accuracy[i]<- round(acc, 4)  
}  
print(results_accuracy)  
cat("Mean Accuracy across" , n_runs , "runs", round(mean(result_accuracy$Accuracy),4), "\n")
```

Glimpse of the data

```
> head(data)
```

```

      id diagnosis radius_mean texture_mean perimeter_mean area_mean smoothness_mean
1  842302         M      17.99      10.38      122.80     1001.0         0.11840
2  842517         M      20.57      17.77      132.90     1326.0         0.08474
3 84300903         M      19.69      21.25      130.00     1203.0         0.10960
4 84348301         M      11.42      20.38       77.58       386.1         0.14250
5 84358402         M      20.29      14.34      135.10     1297.0         0.10030
6  843786         M      12.45      15.70       82.57       477.1         0.12780

compactness_mean concavity_mean concave.points_mean symmetry_mean fractal_dimension_mean
1      0.27760      0.3001      0.14710      0.2419      0.07871
2      0.07864      0.0869      0.07017      0.1812      0.05667
3      0.15990      0.1974      0.12790      0.2069      0.05999
4      0.28390      0.2414      0.10520      0.2597      0.09744
5      0.13280      0.1980      0.10430      0.1809      0.05883
6      0.17000      0.1578      0.08089      0.2087      0.07613

compactness_mean concavity_mean concave.points_mean symmetry_mean fractal_dimension_mean
1      0.27760      0.3001      0.14710      0.2419      0.07871
2      0.07864      0.0869      0.07017      0.1812      0.05667
3      0.15990      0.1974      0.12790      0.2069      0.05999
4      0.28390      0.2414      0.10520      0.2597      0.09744
5      0.13280      0.1980      0.10430      0.1809      0.05883
6      0.17000      0.1578      0.08089      0.2087      0.07613

radius_se texture_se perimeter_se area_se smoothness_se compactness_se concavity_se
1   1.0950   0.9053      8.589 153.40   0.006399      0.04904   0.05373
2   0.5435   0.7339      3.398  74.08   0.005225      0.01308   0.01860
3   0.7456   0.7869      4.585  94.03   0.006150      0.04006   0.03832
4   0.4956   1.1560      3.445  27.23   0.009110      0.07458   0.05661
5   0.7572   0.7813      5.438  94.44   0.011490      0.02461   0.05688
6   0.3345   0.8902      2.217  27.19   0.007510      0.03345   0.03672

area_worst smoothness_worst compactness_worst concavity_worst concave.points_worst
1    2019.0         0.1622         0.6656         0.7119         0.2654
2    1956.0         0.1238         0.1866         0.2416         0.1860
3    1709.0         0.1444         0.4245         0.4504         0.2430
4     567.7         0.2098         0.8663         0.6869         0.2575
5    1575.0         0.1374         0.2050         0.4000         0.1625
6     741.6         0.1791         0.5249         0.5355         0.1741

symmetry_worst fractal_dimension_worst
1      0.4601         0.11890
2      0.2750         0.08902
3      0.3613         0.08758
4      0.6638         0.17300
5      0.2364         0.07678
6      0.3985         0.12440

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